Gel Blot Analyzer v9.8

User Manual

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1. Introduction

1.1 What is Gel Blot Analyzer (GBA)?

Gel Blot Analyzer is a desktop application designed for the analysis of images from gel electrophoresis, Western blots, and similar assays. It provides a comprehensive suite of tools for image annotation, lane quantification, distance quantification, protein and glycan analysis with molecular weight prediction in a single, user-friendly interface.

Key Features

- **Advanced Annotations:** Place standard molecular weight markers automatically or manually, custom text labels, arrows, lines, and rectangles with full control over font, size, and color.
- **Gel Analysis:** Adjust the black point, white point and/or gamma to visualize the protein bands clearly.
- **Powerful Quantification:** Define lane areas using precise quadrilaterals (for skewed lanes) or rectangles. Analyze band intensity profiles with interactive adjustments for peak detection and background subtraction. Results report each band's intensity, **Band %** (purity) and **Lane %**, and **Compare Bands** compares any band of any lane with another. Can pick multiple lane areas for faster quantification and analysis.
- **Standard Curve Analysis:** Generate standard curves from known quantities and automatically calculate the amount of protein in sample lanes.
- **Molecular Weight Prediction:** Predict the molecular weight of unknown bands using standard marker sets.
- **Protein Analysis Tools:** Analyze protein sequences for potential N-glycosylation sites and predict the molecular weight of oligomeric or glycosylated forms, with an option to overlay these predictions on your image.
- **Non-destructive Workflow:** Most image adjustments are previews. Permanent changes like cropping, padding, and rotation are undoable.
- One-Step Auto Analysis: Draw a single boundary around your gel and let the software automatically straighten the image, detect lanes and bands, and run full densitometry through a guided wizard.
- **AI Assistant Control (optional, off by default):** let an AI assistant (Claude, LM Studio, Ollama and others) operate GBA live in your window: inspecting gels, labeling ladders, running densitometry and quantifying against your standards (see 10.6).
- Measurement & Calibration Tools: Calibrate the image to a known length and measure distances and areas directly on the image in real-world units.
- **Flexible Export:** Save your work as a high-resolution image with annotations, a separate configuration file, or export quantitative data directly to Microsoft Excel. Keeps the transparency of the images for posters or other illustrations.
- **Autosave & Crash Recovery:** A recovery copy of your image and all of its work is saved in the background. If the program closes unexpectedly, the next launch offers to reopen the last file with every change restored.

2. Getting Started

2.1 Installation & Dependencies

Gel Blot Analyzer is a Python-based application. There are multiple ways to run the application. You can download the compiled application. However, since the application is not signed so running it on Windows or MacOS will show an error that this app is from an unknown publisher. As long as you downloaded it from GitHub Releases, the application is tested and safe to use. However, if you want to run the code and have programming skills, you can compile the application or run it directly as well.

Option 1: Download the Pre-built Application (Recommended)

The easiest way to get started is to download the latest pre-built application for your operating system.

1. Go to the **Releases** page of this repository.
2. Download the newest release based on numbering:
   - **For macOS:** Download the .zip file.
   - **For Windows:** Download the .zip file.
3. Unzip the file and run the application. No installation is needed.

NOTE: Since the application for MacOS is not signed it may not run or show damaged application, so copy the .app file to the Applications Folder, open terminal and run this command **xattr -cr "/Applications/Gel Blot Analyzer.app"**. For Windows, since it is not from a popular publisher it may show an error that it is blocked by Windows Smart Screen (Windows Defender) so open the application and click on More Info and click on Run Anyway.

Option 2: Run from Source (For Coders)

If you are a coder and want to run the script directly, you can use Conda for a stable environment.

Using Anaconda (Recommended for Stability)

1. **Clone the Repository/Download the Library:**

   ```
   git clone https://github.com/Anindya-Karmaker/Gel-Blot-Analyzer.git
   cd Gel-Blot-Analyzer
   ```

   Or you can download the whole repository as .zip file by clicking on the green button named **<> Code** and select **Download Zip** on the main GitHub Page

2. **Create a New Conda Environment:** This command uses the provided file to create a new environment named **gel_analyzer_env** with all the correct library versions.

   conda env create -f environment.yml

3. **Activate the Environment:**

   conda activate gel_analyzer_env

4. **Run the Program:**

   python Gel_blot_analyzer.py
   NOTE: If you get any error or crashes particularly due to missing library or C-code header mismatch (usually happens on Windows- Anaconda library issue), run this code:
   pip install --force-reinstall --no-cache-dir numpy scikit-image

Using pip

You can also use pip with the provided requirements.txt file, although dependency conflicts can be more common.

1. **Clone the Repository/Download the Library:**

   git clone https://github.com/Anindya-Karmaker/Gel-Blot-Analyzer.git
   cd Gel-Blot-Analyzer

   Or you can download the whole repository as .zip file by clicking on the green button named **<> Code** and select **Download Zip** on the main GitHub Page.

2. **Create and activate a virtual environment (Recommended):**

   python -m venv venv
   # On Windows
   .\venv\Scripts\activate
   # On macOS/Linux
   source venv/bin/activate

3. **Install requirements:**

   pip install -r requirements.txt

4. **Run the program:**

python Gel_blot_analyzer.py

NOTE: If you get any error or crashes particularly due to missing library or C-code header mismatch (usually happens on Windows- Anaconda library issue), run this code:

pip install --force-reinstall --no-cache-dir numpy scikit-image

2.2 Building the Application from Source

If you wish to compile the application yourself, you will need pyinstaller.

1. **Activate your Python environment** (either Conda or venv) where you have installed the dependencies.

2. **Install pyinstaller:**

   pip install pyinstaller

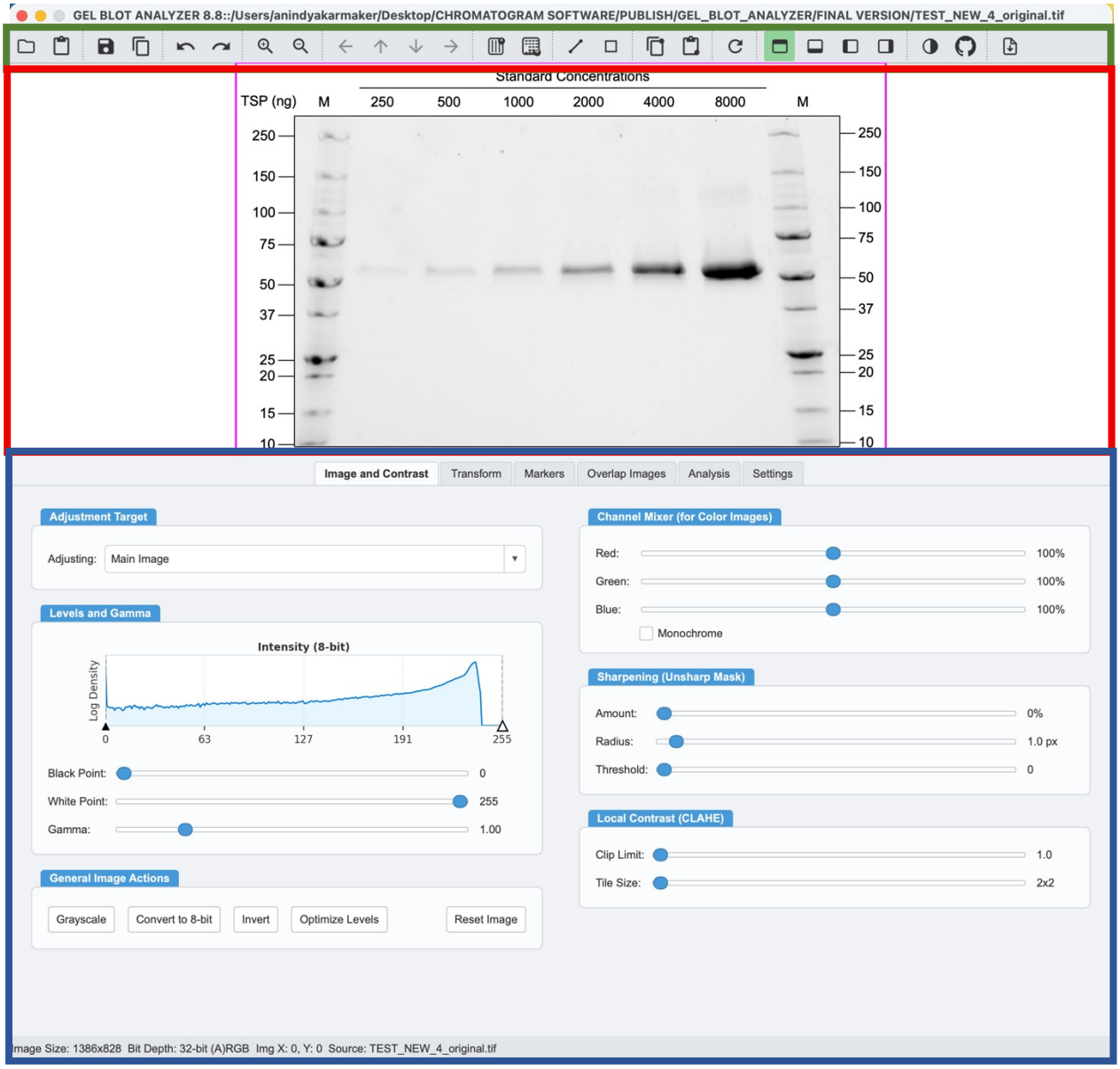
3. **Run the Build Command:** This project includes pre-configured .spec files for both macOS and Windows. Using these is the most reliable way to build.

   - **On macOS:** pyinstaller build_macos.spec
   - **On Windows:** pyinstaller build_windows.spec

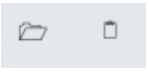
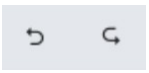
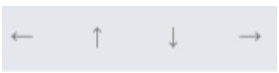
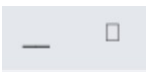
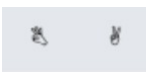
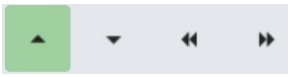
   The final, self-contained application will be located in the **dist** folder.

2.3 The Main Interface

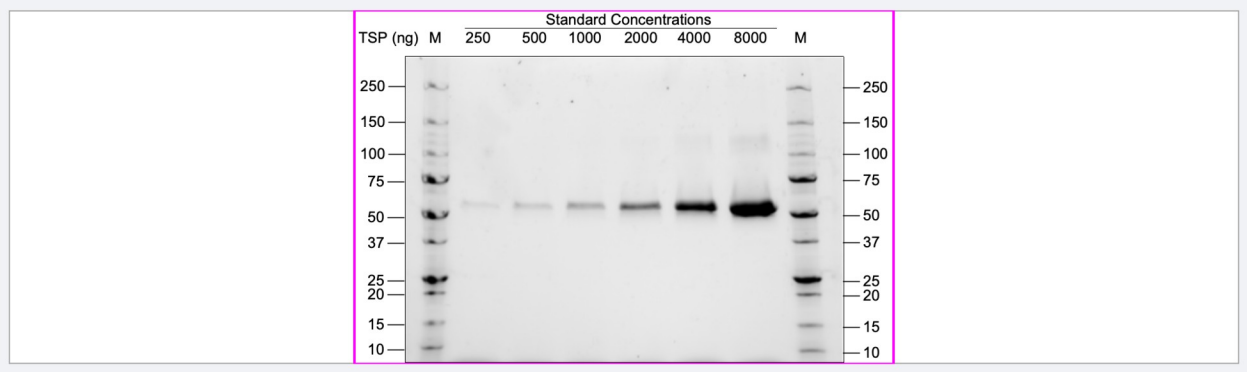
The main window is divided into three primary sections: Toolbar, Image Viewer, and Image Control/Adjustment Tabs.



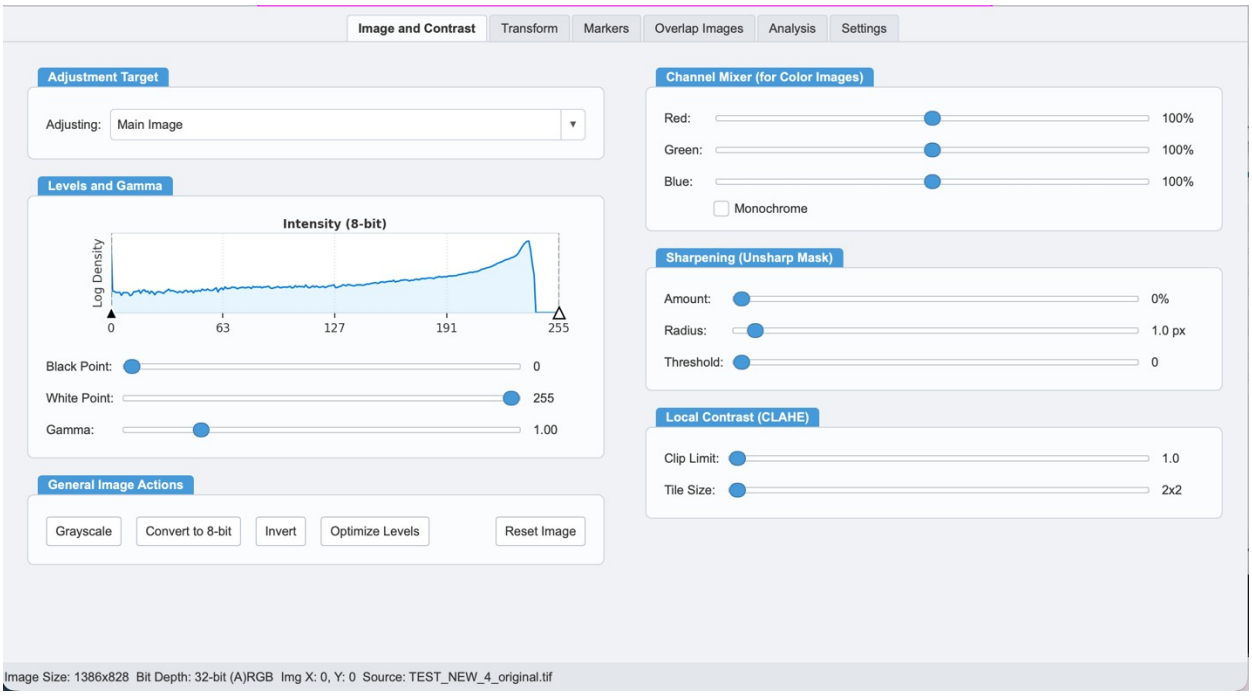
1. **Toolbar:** Provides quick access to the most common functions like loading/saving files, undo/redo, zoom, pan, and annotation tools.

Functions	Icons (v8.0 onwards)	Icons (v7.0)
Open, Paste from Clipboard		
Save, Copy to Clipboard		
Undo, Redo		
Zoom In, Zoom Out		
Pan Left, Pan Top, Pan Bottom, Pan Right		
Automatic Lane Marker Labeling		
Auto Gel Analysis		
Create Line, Create Rectangle		
Copy Marker Layout, Paste Marker Layout		
Reset Changes		
Image Viewer Location: Top (Selected), Bottom, Left or Right		
Dark/Light Mode		
Open GitHub Link		
Save as Microsoft® PowerPoint		

2. **Image Viewer:** The central area where your image is displayed. This is an interactive canvas where you will place markers, draw shapes, and define analysis regions. You can change the location of the Image Viewer by clicking on the Image Viewer Location icons. The **magenta** boundary box indicates the rendering region which will be exported when you save image or copy the image to clipboard.



3. **Settings Tabs:** A collection of tabs that organize all the detailed controls for Image and Contrast adjustment, Transform and pad images, Markers placement, Overlap Images (for blending/combining different images), Analysis for densitometric and protein analysis, and a Settings tab for application preferences such as CPU/GPU processing, UI scaling, and window dimensions.



3. Core Concepts

3.1 The Undo/Redo System

Nearly every action that modifies the image or its annotations (e.g., adding a marker, cropping, applying padding, rotating) is saved to an undo stack. It is possible to go back up to 20 steps. Image contrast settings can be reset/adjusted infinitely.

Undo: Use the undo button on the toolbar or the shortcut (Ctrl+Z / Cmd+Z) to reverse the last action.

Redo: Use the redo button or shortcut (Ctrl+Y / Cmd+Shift+Z) to redo the action.

3.2 Saving Your Work

The software uses a paired-file saving system to preserve all your work, including annotations and settings. When you save, it creates three files:

- **[basename]_original.png**: A copy of the modified image *before* any annotations were added.
- **[basename]_modified.png**: A high-resolution image with all your markers and annotations "burned in".
- **[basename]_config.txt**: A text file containing all the data for your markers, shapes, and settings.

When you load the _original.png file, the software will automatically find and apply the corresponding _config.txt file, restoring your session exactly as you left it. Even if you copy the image and paste it, based on the labeling of the file the configuration will be loaded and applied.

Note: Please do not change the file name by removing **_original** or **_config** otherwise the program will fail to load the files properly.

3.3 Autosave and Crash Recovery

GBA keeps a private recovery copy of the open image together with everything you have done to it: labels, markers, custom text and shapes, adjustments, analysis regions and densitometry results. Only changes are saved, so an unchanged file costs nothing, and your own files are never overwritten. The status bar shows **Autosaved** and the time of the last recovery copy.

If the program closes unexpectedly, the next launch opens a **Recover Unsaved Work** window showing a preview of the unsaved image and when it was saved. Click **Restore** to reopen it with every change applied, **Discard** to delete the recovery copy, or **Decide Later** to keep it and be asked again next time. A restored image is marked **(Recovered)** in the title bar; save it to keep the changes.

Recovery copies are deleted automatically when you close the program normally. Set how often they are saved, or switch autosave off, in **Settings ▸ Autosave & Crash Recovery** (see 10.5).

4. Basic Operations: File & View

4.1 Loading an Image

Action: Click the "Load Image" button on the toolbar or use the shortcut (Ctrl+O / Cmd+O).
Function: Opens a dialog to select an image file (.png, .jpg, .tif, etc.). If a corresponding _config.txt file is found, it will be loaded automatically. You can load even raw 16-bit images but the raw images may need further adjustments like image invert operation and adjusting the black and white point values in the **Image and Contrast** tab.

4.2 Saving an Image

Action: Click the "Save with Config" button or use the shortcut (Ctrl+S / Cmd+S).
Function: Opens a dialog to save your work using the paired-file system described in Core Concepts. It is possible to save the image using different formats (PNG, JPG, TIFF, etc). PNG is ideal for pasting into presentations or documents where transparency is necessary and TIFF is ideal for lossless images ideal for publications.
Saving as JPG/JPEG: JPEG cannot store transparency and is lossy. When you choose a .jpg or .jpeg file name, GBA warns you before saving: transparent areas such as padding would be filled with white permanently, and band intensities are slightly altered on every save, which affects densitometry. Choose **Switch to PNG** or **Switch to TIFF** (both lossless, both keep transparency), **Keep JPEG**, or **Cancel**.
Save to PowerPoint (.pptx): Use **Save to PPTX…** (on the toolbar and in the **File** menu) to export the figure as an editable PowerPoint file. The gel/blot image goes in as a single high-resolution background picture, and every annotation — the Left/Right/Top markers, custom text markers, and custom shapes (lines, rectangles and arrows) — is added as a native, fully editable PowerPoint text box or shape. You can click any label in PowerPoint to change its text, font, size or colour (or copy the whole figure from PowerPoint into Word). This is ideal when you need to tweak annotations after export without re-rendering in GBA. Each text box is sized to fit its text closely, like the label boxes in GBA, so labels are easy to select and edit in PowerPoint.

4.3 Copy & Paste

Copy Image (Ctrl+C / Cmd+C): Copies the currently displayed image *with all visible annotations* to the clipboard as a high-resolution PNG in the clipboard for pasting directly into other software like Microsoft® Word or PowerPoint etc.
Paste Image (Ctrl+V / Cmd+V): Pastes an image from the clipboard into the viewer, replacing the current image. If the clipboard contains a path to an image file, it will load that file and its associated config if available.

- A pasted image has no file of its own, so **Save** asks you for a new name and folder rather than suggesting the previously opened file. If the clipboard is empty, GBA tells you so.

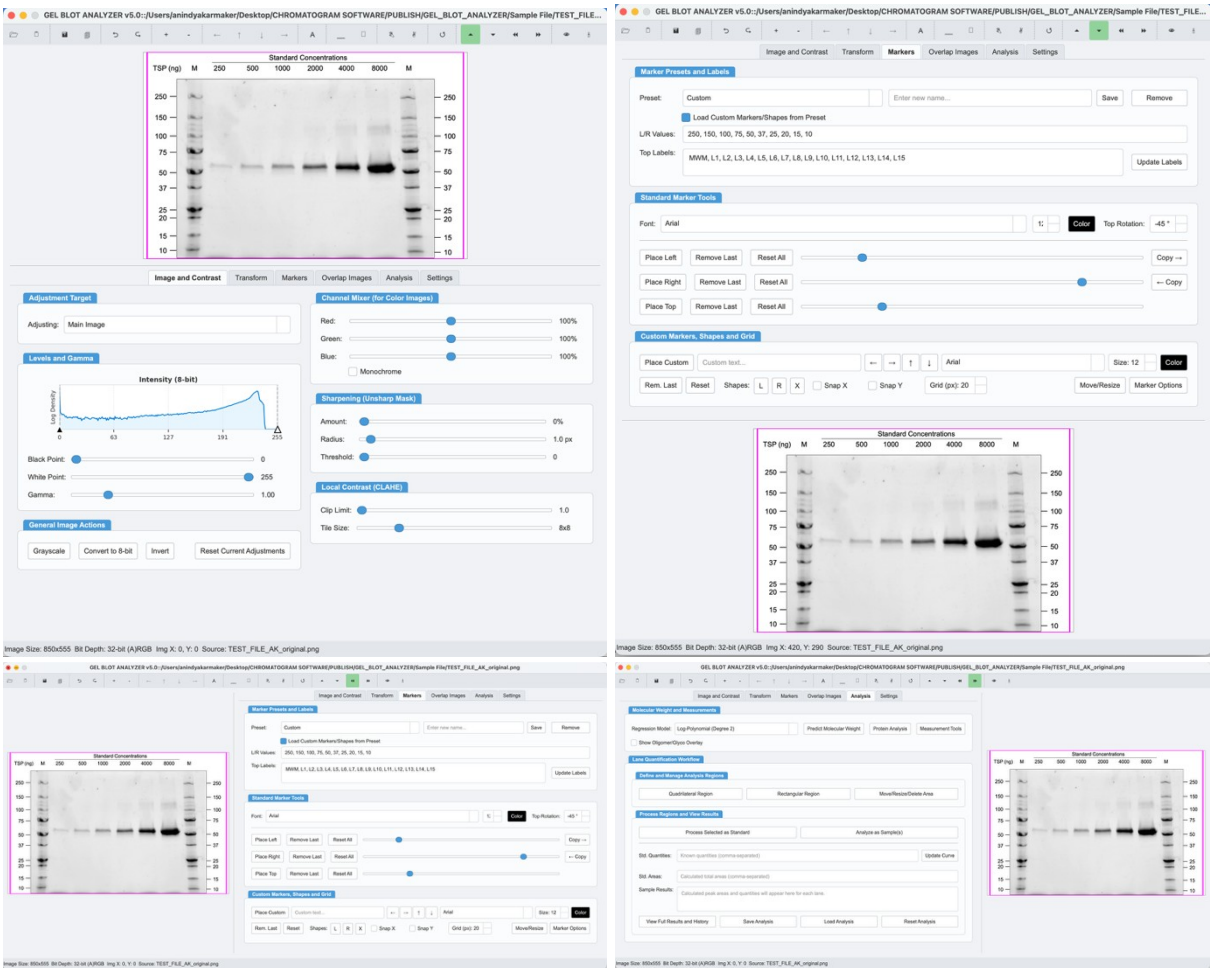
4.4 Navigating the Image: Zoom & Pan

Zoom In/Out:
Use the + and - buttons on the toolbar.
Use the mouse wheel or 2-finger pinch on a touchpad to zoom in or out the image.
Use shortcuts (**Ctrl** + **=** or **Cmd** + **=** for zoom in, **Ctrl** + **-** / **Cmd** + **-** for zoom out).

Pan (Move the View):
When zoomed in, **click and drag** the image with the **right mouse** button.
Use the arrow buttons on the toolbar or use the arrow keys on your keyboard.

4.5 Changing the Layout

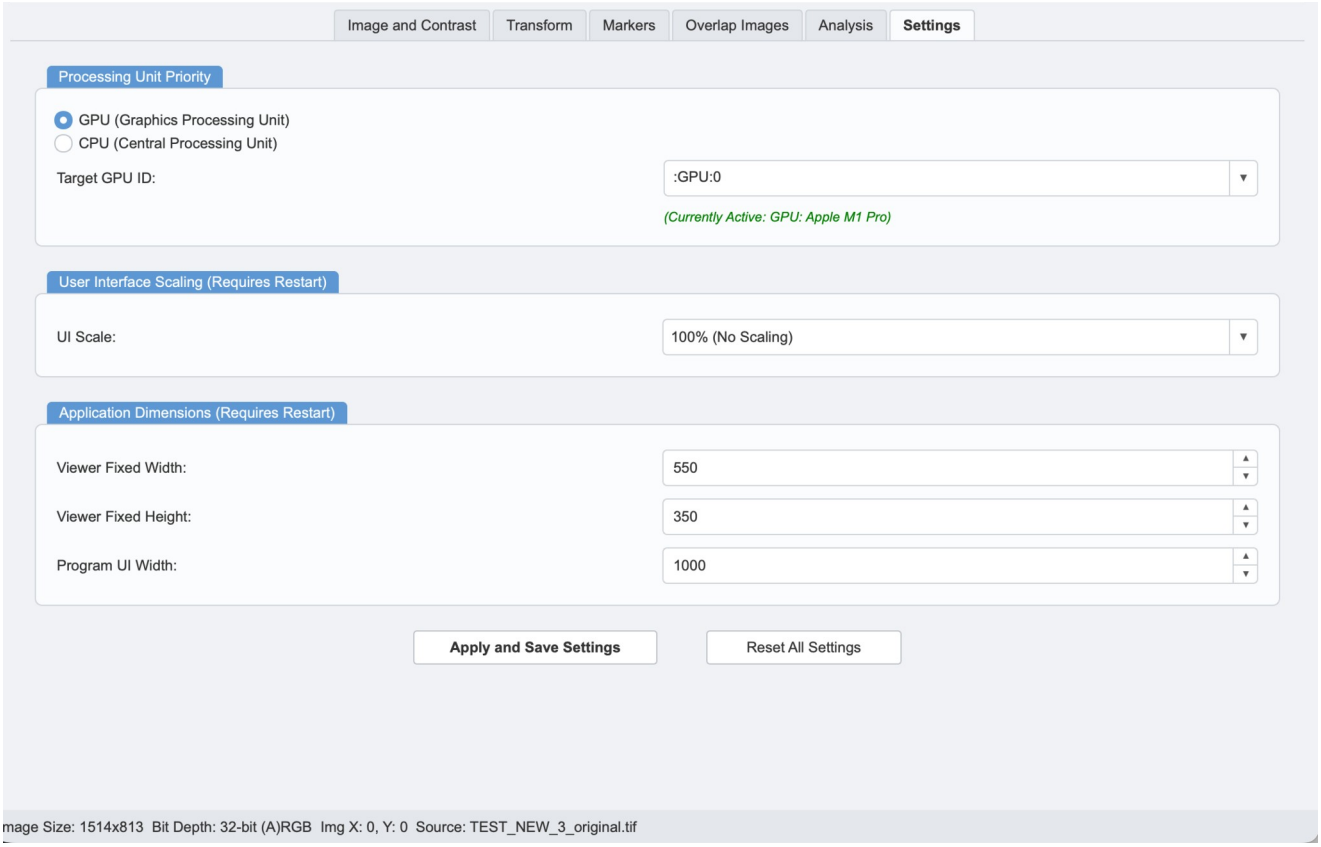
You can change the position of the Image Viewer relative to the Settings Tabs for your convenience. Use the layout buttons on the toolbar to position the viewer on the Top, Bottom, Left, or Right.



4.6 Adjusting the User Interface

You can control the size of the Image Viewer Area (Viewer Fixed Width/Viewer Fixed Height) and the width of the control section using the Program UI width. Moreover, if the icons/symbols appear too or small, please adjust the UI scaling from 0% to 250% from the Settings Tab. Note: Some settings require restart of the application to work properly. The application by default uses GPU for acceleration but it can be toggled off to use mostly CPU.

Resizing the Top Labels and Sample Results boxes: drag the grip in the bottom-right corner of either box to make it taller or shorter. Double-click the grip to restore the default height. The height you choose is remembered the next time you open the program.


Image and Contrast
Transform
Markers
Overlap Images
Analysis
Settings
Processing Unit Priority
GPU (Graphics Processing Unit)
CPU (Central Processing Unit)
Target GPU ID:
:GPU:0
(Currently Active: GPU: Apple M1 Pro)
User Interface Scaling (Requires Restart)
UI Scale:
100% (No Scaling)
Application Dimensions (Requires Restart)
Viewer Fixed Width:
550
Viewer Fixed Height:
350
Program UI Width:
1000
Apply and Save Settings
Reset All Settings
mage Size: 1514x813 Bit Depth: 32-bit (A)RGB Img X: 0, Y: 0 Source: TEST_NEW_3_original.tif

5. Image Adjustments & Transformations

5.1 Image and Contrast tab

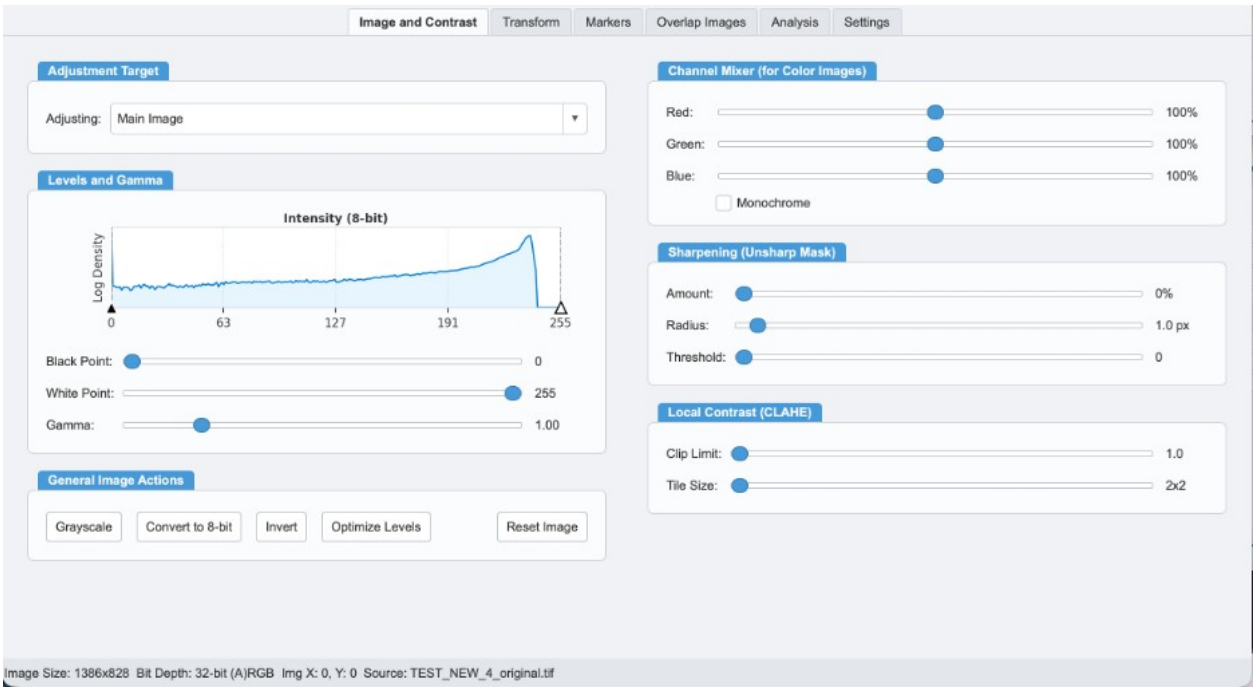
The **Image and Contrast** tab provides a suite of tools to optimize the visual appearance of the image for analysis. With the exception of "Grayscale", “Convert to 8-bit” and "Invert," all adjustments are non-destructive previews, meaning they do not permanently alter the original image data until you perform an action like rasterizing.

Adjustment Target: This dropdown menu is the most important control on this tab. It selects which image is currently being adjusted: the **Main Image**, or one of the overlays (**Overlay 1** or **Overlay 2**). This allows you to apply different contrast, color, and sharpening settings to each image independently, which is essential for matching the appearance of different images (e.g., a gel and a western blot) when creating a composite overlay (for more details see **Overlap Images** tab).

Levels and Gamma: These sliders control the overall brightness and contrast.

Black/White Point: Adjusts the dynamic range. Pixels darker than the **Black Point** become pure black, and pixels lighter than the **White Point** become pure white. This is the primary tool for maximizing contrast. The effect is visualized by the markers on the **Log-Histogram** window.

Gamma: Adjusts the brightness of the image's mid-tones without affecting the absolute black and white points, helping to reveal faint details.



Channel Mixer (for Color Images): This tool gives you precise control over the color channels in an RGB image. Increase or decrease the contribution of each color channel (RGB). This is useful for isolating signals from specific fluorescent channels or for creating a custom color balance. Monochrome: Creates a grayscale image based on the weighted contributions of the red, green, and blue channels that have been set.

Sharpening (Unsharp Mask): Increases the contrast along the edges of features, making them appear sharper. Amount: Controls the strength of the sharpening effect. Radius: Determines the size of the area around the edges that will be affected. Threshold: Prevents sharpening in low-contrast areas, which helps to avoid amplifying image noise.

Local Contrast (CLAHE): Contrast **L**imited **A**daptive **H**istogram **E**qualization improves contrast by enhancing it locally across different regions of the image, rather than globally. This is excellent for observing or visualizing very faint bands without causing saturation of other bands. Clip Limit: Controls how much contrast is amplified, preventing noise from being excessively brightened. Tile Size: Defines the size of the local regions for histogram equalization. Higher is better.

General Image Actions:

Grayscale (Ctrl+B / Cmd+B): A permanent action that converts the image to a high-precision grayscale format, which is recommended for most quantitative analyses. Can use Undo option to revert back the changes.

Convert to 8-bit: If you program is running really slow and you do not need increased precision, this option can lower the image resolution for easier operation. However, quality may be affected. Can use Undo option to revert back the changes.

Invert (Ctrl+I / Cmd+I): A permanent action that inverts all image colors (e.g., black to white). This is essential for densitometry specially for DNA/RNA gel analysis, as it transforms dark bands into positive peaks for analysis. Can use Undo option to revert back the changes.

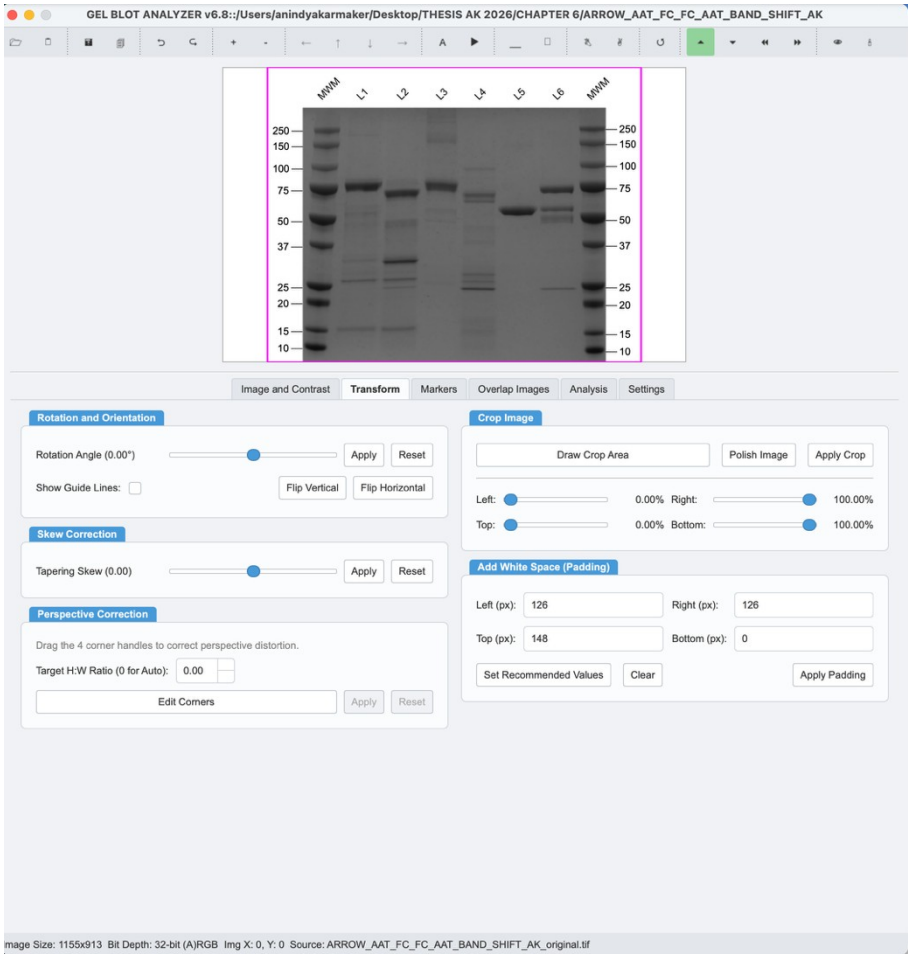
Optimize Levels: Auto adjusts the black and white point for optimum display and analysis. May not work with some images with multiple peaks.

Reset Image: Resets every control on this tab — Levels/Gamma, Channel Mixer, Sharpening and Local Contrast (CLAHE) — to its default for the currently selected adjustment target. In a narrow window the buttons in this group wrap onto a second row, so their labels are never cut off.

NOTE: Densitometric analysis applying most image or contrast options particularly **Channel Mixer or CLAHE** may change the band analysis significantly and user caution is advised.

5.2 Transform tab

The **Transform** tab contains all the tools needed to geometrically correct and prepare your image for analysis. It is organized into two main sections: non-destructive **Global Transformations** that can be viewed as live previews and then applied, and destructive **Cropping and Padding** operations that permanently alter the image canvas (can undo changes).



5.3 Global Transformations (Live Previews)

These tools allow you to see the effect of your changes in real-time before applying them.

Rotation and Orientation: Use the **Rotation Angle** slider for a live preview to straighten a tilted gel or blot. You can toggle the **Show Guide Lines** checkbox (Shortcut: Ctrl+G) to display a crosshair, making precise alignment easier. Once you are satisfied, click **Apply** to make the rotation permanent. Note that applying a rotation will remove any standard L/R/Top markers, as their alignment becomes ambiguous. The **Flip** buttons provide immediate, permanent vertical or horizontal mirroring of the image.

Skew Correction: This tool corrects for perspective distortion, often seen as "smiling" or "frowning" bands in a gel. Use the **Skew** slider for a live preview of the correction. When the lanes appear parallel, click **Apply** to make the transformation permanent.

5.4 Destructive Edits (Permanent Actions)

These actions modify the fundamental dimensions of your image. It is recommended to perform these edits before adding detailed annotations.

Crop Image: This is used to remove unwanted areas of the image.

- **Draw Crop Area:** The primary method. Click this button, then click and drag on the image to define the area you want to keep.
- **Sliders:** For precise adjustments, you can use the sliders to define the crop boundaries as a percentage of the image dimensions.
- **Apply Crop:** This is the final step. Clicking this button permanently removes everything outside the defined crop area. All existing markers and shapes will be repositioned relative to the new top-left corner.

Add White Space (Padding): This permanently adds a blank border around your image. This is extremely useful for creating clean space for markers and other annotations, ensuring they don't cover important image data. You will notice the changes as the magenta boundary region will change highlighting the change in the rendered output.

- **Input Fields:** Enter the number of pixels to add to each side.
- **Set Recommended Values:** A convenient button that suggests default padding values based on the current image size.
- **Apply Padding:** Click this to finalize the action. The image canvas will be enlarged, and all existing markers and shapes will be automatically shifted to maintain their correct positions relative to the original image content.

Perspective Correction: The user picks four corner points based on the level of skewness of the gel/image and it can auto-adjust the final image into a perfect rectangular shape with the perspective corrected.

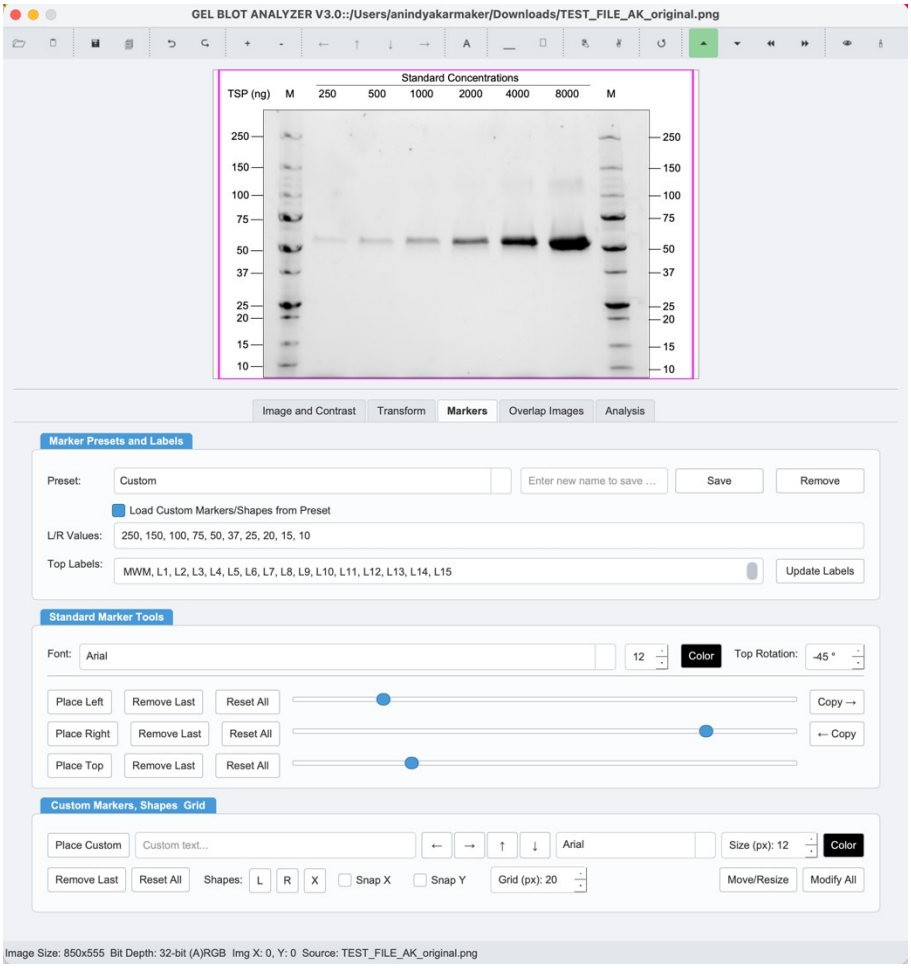
- **Target H:W ratio**: Default is set to auto (0) but the user can make it can fix height to width ratio, useful if you know the gel height to width ratio.

Note: Perspective correction and/or skew correction modifies the bands significantly. This feature is significantly useful to get accurate molecular weight marker prediction, but for densitometry, it can change the values. User caution advised.

6. Annotations: Markers & Shapes

6.1 Markers tab

The **Markers** tab is the control center for all annotations.



Standard Markers (Left, Right, Top)

These are typically used for molecular weight standards or lane labels.

1. **Select a Preset:** Choose a standard from the **Preset** dropdown (e.g., a Bio-Rad or Thermo ladder). This populates the internal list of values. If you want to create your own markers then choose the **Preset** that is closest to your target and then choose the **Preset: Custom** and enter or edit the values separated by comma (,) and give a suitable name and click on **Save Preset**. The save preset function also saves the **Top Marker Labels** as well. The **Save Preset** will also save any **Custom Markers** as well. If you load a **Preset** but do not want to load the saved **Custom Markers**, please uncheck the option **Load Custom Markers/Shapes from Preset.**
2. Place Markers:
   - Click **Place Left** (Shortcut: Ctrl+Shift+L / Cmd+Shift+L).

- Click on the image at the vertical position of each corresponding marker band, from top to bottom. The software will automatically apply the labels from the preset.
- Repeat for **Place Right** (Ctrl+Shift+R / Cmd+Shift+R) and **Place Top** (Ctrl+Shift+T / Cmd+Shift+T).

3. **Copy Markers**: If you want to duplicate the markers from left side or right side, click the (Copy →) or (Copy ←) button.
4. **Adjust Position:** The **Left position (X)**, **Right position (X)** and **Top position (Y)** sliders move all the labels on one side together; the number next to each slider is that position in image pixels. Use them if image editing has pushed the markers out of the preview window.
5. **Drag labels in any direction:** In **Move/Resize** mode (Ctrl+Shift+M / Cmd+Shift+M), drag a left or right label up/down to move only that label, or left/right to move the whole column — its position slider follows. Top labels work the other way round: left/right moves one label, up/down moves the whole row. The arrow keys do the same (hold Shift for 10 px steps).
6. **Remove/Reset:** Use the **Remove Last** to remove last marker entry or **Reset All** buttons to reset the markers.

You can also modify the font type, size, color and rotation (for top labels) by changing the variables in this tab.

NOTE: If **Line X** or **Line Y** is on along with the **Snap**, the markers and other parameters will be constrained along the center of the grid lines.

6.2 Custom Markers & Shapes

Use this section for any other annotations like placing arrows or molecular weight, etc.

1. **Custom Text/Arrows:**
   - Enter text in the text box or click an arrow button from the default presets.
   - Select the desired font, size, and color.
   - Click Place Custom.
   - Click on the image to place the marker.

Note: You can change the color, text, size and font of the custom markers and can see an overlay before placing it. Please click **Place Custom** if you change the parameters to see the updated overlay of the custom text/shape.

2. **Shapes (Lines, Rectangles & Arrows):**
   - Select the desired size (thickness) and color.
   - Click the **L** (Line), **R** (Rectangle) or **A** (Arrow) button in the “Shapes” section.
   - Click and drag on the image to draw the shape. Hold **Shift** while drawing to constrain lines to 45° angles and rectangles to squares.
   - **Arrows:** The **A** tool draws an arrow in any direction — the arrowhead points to wherever you release the mouse. Hold **Shift** to constrain the angle to 0°, 45° or 90°. In **Move/Resize** mode, drag the two end handles to re-aim or lengthen the

arrow, and drag the two side handles (perpendicular to the shaft) to make it thicker or thinner.

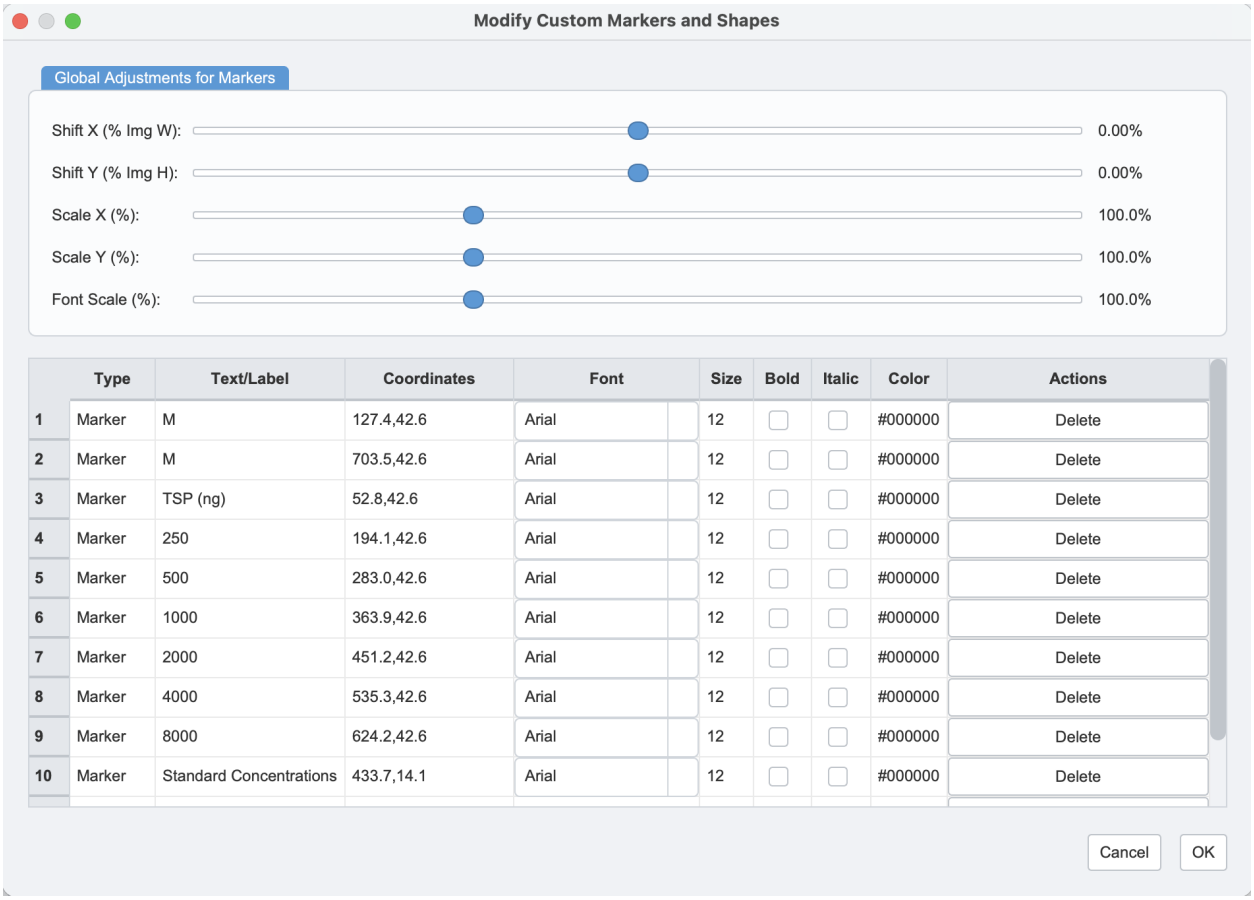
3. **Move/Resize Custom Items:** Click the **Move/Resize** button (or press **Ctrl+Shift+M** / **Cmd+Shift+M**; press it again or Esc to exit) to enter interaction mode. You can then click and drag markers or shapes to reposition them; for shapes, drag the corner handles to resize. Double-click any text or shape to edit its text, font, color, thickness or coordinates.

Moving several items at once: in Move/Resize mode, **Shift+click** custom markers and shapes to add them to (or remove them from) the selection; selected items show a dotted magenta outline. You can release Shift and then drag any selected item to move the whole group, just like in PowerPoint. The arrow keys nudge the whole group (hold Shift for 10 px steps), and Delete removes all selected items. Clicking an item that is not selected, or leaving Move/Resize mode (Esc), clears the selection.

NOTE: If **Line X** or **Line Y** is on along with the **Snap**, the markers/shapes will be constrained along the center of the grid lines.

6.3 Managing Custom Markers/Annotations

You can use the **Marker Options** to open the **Modify Custom Markers and Shapes** window to adjust parameters like Text/Label, Location, Font Type, Font Size and other properties.



	Type	Text/Label	Coordinates	Font	Size	Bold	Italic	Color	Actions
1	Marker	M	127.4,42.6	Arial	12	☐	☐	#000000	Delete
2	Marker	M	703.5,42.6	Arial	12	☐	☐	#000000	Delete
3	Marker	TSP (ng)	52.8,42.6	Arial	12	☐	☐	#000000	Delete
4	Marker	250	194.1,42.6	Arial	12	☐	☐	#000000	Delete
5	Marker	500	283.0,42.6	Arial	12	☐	☐	#000000	Delete
6	Marker	1000	363.9,42.6	Arial	12	☐	☐	#000000	Delete
7	Marker	2000	451.2,42.6	Arial	12	☐	☐	#000000	Delete
8	Marker	4000	535.3,42.6	Arial	12	☐	☐	#000000	Delete
9	Marker	8000	624.2,42.6	Arial	12	☐	☐	#000000	Delete
10	Marker	Standard Concentrations	433.7,14.1	Arial	12	☐	☐	#000000	Delete

The **Shift X** and **Shift Y** shift the overall markers X and Y axis respectively. The **Scale X** and **Scale Y** shrink or expand the custom markers. The **Font Scale** adjusts all of the custom marker fonts globally except the Top/Left/Right markers since they have their own font/color and other properties settings.

Copy/Paste Custom Items (Experimental Feature): Use the copy/paste buttons on the toolbar to copy all custom markers and shapes from one image, load the other image and then paste them onto another. Use the Shift X/Y to change the location of the pasted markers and Scale X/Y to control their size. Font Scale changes the global font size for everything.

Note: Only use the copy/paste after placing the left/right/top markers otherwise the custom markers will be reset.

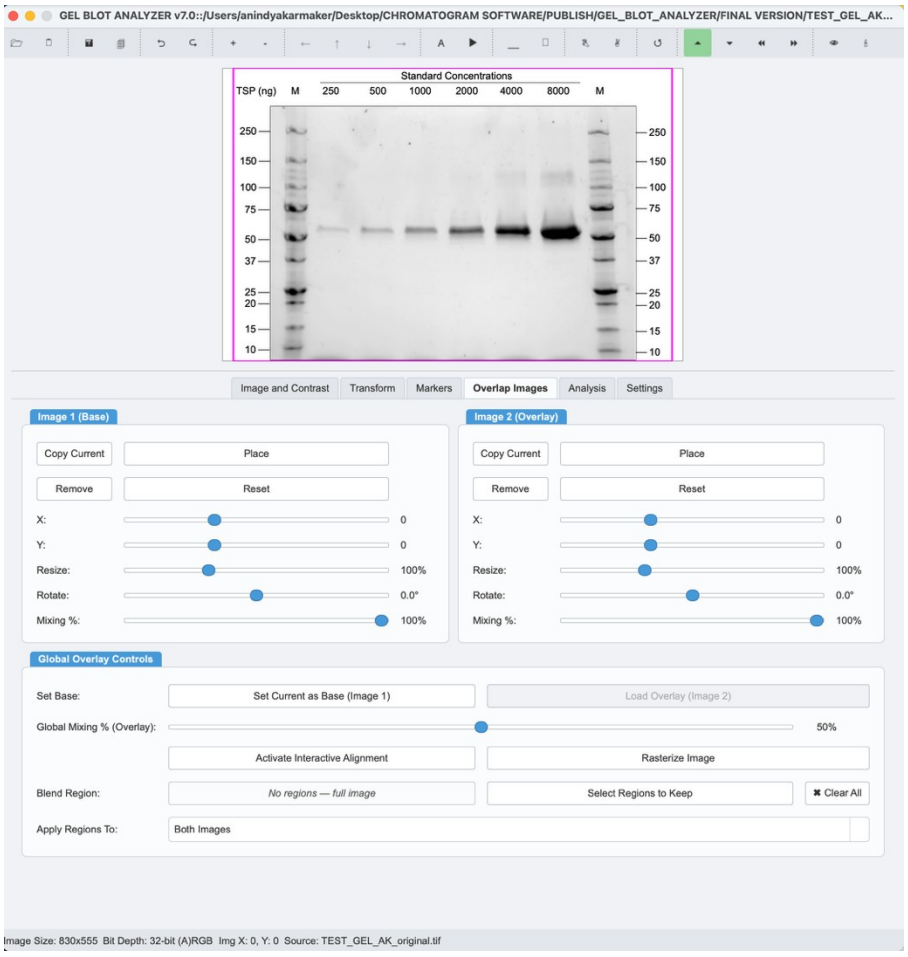
6.4 The Grid System & Snapping

- **Show Grid:** Check **Line X** (Shortcut: Ctrl+Shift+X or Cmd+Shift+X) and **Line Y** (Shortcut: Ctrl+Shift+Y or Cmd+Shift+Y) to display a grid.

- **Grid Size:** Adjust the grid spacing in pixels using the **Grid (px)** spinbox (Shortcut: Ctrl+Shift+G or Cmd+Shift+G to enable the grid). To increase or decrease the grid size, use the **Grid (px)**: spinbox (Shortcut: Ctrl+Shift+ ↑ or Cmd+Shift+ ↑ and Ctrl+Shift+ ↓ or Cmd+Shift+ ↓)
- **Snap:** When enabled (Shortcut: Ctrl+Shift+N or Cmd+Shift+N), every mouse click — placing markers, drawing shapes, etc. — snaps to the nearest grid intersection. Turning Snap on automatically shows the grid so you can see what you are snapping to: if neither Line X nor Line Y is currently displayed, both are switched on; if one is already on, your choice is kept and snapping simply uses it. Turning Snap off leaves the grid lines as they are.

7. Overlap Images

You can overlap a WB on top of the SDS-PAGE or combine multiple images using this function.



1. Paste an image into the image viewer and make any edits using the previous tabs. Once you are happy with the changes, click on **Copy Current Image** in the **Image 1 (Base)** section and it will copy it into memory.

2. Load the second image if you have a second image, make the changes and click on **Copy Current Image** in the **Image 2 (Overlay)** section.
3. Now load the final or last image on top of which you want to place the other two images. Use the **Padding** tab to create the required spacing to place the other two images or if you want to overlap then on top of each other.
4. Click on **Place** in the **Image 1 (Base)** section and/or **Place** in the **Image 2 (Overlay)** section. Place adds the layer on top of the image you currently have open — it does not replace it — so you can build up a figure from several gels and blots. (The one exception is **Set Current Image as Base** below: because that makes Image 1 a copy of the image already on screen, the copy is shown instead of the original rather than on top of it, so it is not drawn twice.)
5. You can use the sliders to adjust the size, position, rotation or level of blending. You can click on **Activate Interactive Overlay Mode** and you can use it to resize or move the image by dragging with mouse.
6. **Remove** takes a layer off the canvas but keeps it loaded, so you can reposition or re-stack it and press **Place** again without copying or loading it a second time. Use **Reset** when you actually want to empty the Image 1 or Image 2 slot — **Reset** is the only control that discards the stored image.
7. Once you are happy with the changes, click on **Rasterize Image** to finalize it. Markers and labels are never burned into the picture — they stay editable. Rasterizing takes the placed layers off the canvas (they are now part of the image itself) but leaves them loaded in their slots, so you can immediately **Place** the next panel against what you just baked in and rasterize again — repeat this to assemble a multi-panel figure. Rasterizing also ends **Interactive Alignment** and resets Levels/Gamma and the other **Image and Contrast** settings (for the main image and for the loaded slots), so the finished figure starts from neutral settings. **NOTE:** Rasterizing bakes the layers permanently into the image. It is a single undo step, so Ctrl+Z (Cmd+Z) puts the layers back exactly as they were placed — but once you save or move on, the merge cannot be taken apart again.
8. If you just want to merge two images like Chemiluminescence and Colorimetric images for WB (does not need individual adjustments), you can load the first image and click on **Set Current Image as Base (Image 1)** and then click on **Load Overlay (Image 2)** to load the second image.
9. You can use **Blend Region** to create selection rectangle to keep areas that you want to blend/overlap and use the **Apply Regions To** select which areas need to be blended.

Note: Default blending is set to 50% for global and 100% for individual. If you want to overlap images on-top this behavior is perfect. However, if you want overlap one image in an empty spot on another image, please set the blending to 100%.

Overlap Images — what each control does

Each of the two panels, **Image 1 (Base)** and **Image 2 (Overlay)**, has the same set of controls:

Copy Current — copies the image currently open into that slot's buffer. It does not put anything on the canvas yet; the image is simply held, ready to Place. The snapshot is taken as the image is, so a Rotation Angle you have dialled in but not yet applied is NOT baked into it. The copy keeps the image exactly as it is, including any transparent padding (it is no longer trimmed). It also remembers that image's markers and labels — custom markers, shapes, left/right MW labels and top labels — so they can be brought along when you place the panel on another image.

Place — puts the buffered image onto the canvas at the slot's X/Y position. It is added on top of the image you currently have open rather than replacing it, so you can build a figure from several gels and blots. (Set Current as Base is the exception — see below.) A layer is clipped to the image's own edges on all four sides: anything you push beyond them is not shown, because Rasterize and Save would discard it. Add padding in the Transform tab to make room first. If the panel was copied from a different image, Place first asks which of its markers and labels to paste (custom markers/text, lines/arrows/rectangles, left/right MW labels, top labels). Every box starts unticked, and Cancel does not place the panel. Ticked items are added as editable markers when you Rasterize.

Remove — takes the layer off the canvas but keeps it loaded in the slot, so you can reposition or re-stack it and press Place again without copying or loading it a second time.

Reset — empties the slot completely, discarding the stored image. This is the only control that throws the image away; use Remove if you only want it off the canvas.

X / Y — the layer's position in image pixels. Negative values move it off the left/top edge.

Resize — scales the layer. Arrow keys nudge it in 0.01% steps once the slider has focus, which is how you match a layer to the image underneath.

Rotate — turns that layer alone. Rotating the main image in the Transform tab does not affect placed layers; this is the control for turning one.

Mixing % — that layer's own opacity. Image 1 uses this value alone; Image 2 is multiplied by the Global Mixing % as well.

The **Global Overlay Controls** below them apply to the composition as a whole:

Set Current as Base (Image 1) — copies the current image into Image 1 and places it in one step. Because Image 1 is then a duplicate of the image already open, it is shown INSTEAD of it rather than on top of it, so the same picture is not composited twice.

Load Overlay (Image 2) — loads a second image from disk into Image 2. The file dialog opens in the folder the current image came from.

Global Mixing % (Overlay) — applies to Image 2 only, on top of its own Mixing %. Image 1 ignores it. The default is 50%, which suits overlaying two exposures of the same blot; set it to 100% when you are placing a separate panel into empty space.

Activate Interactive Alignment — click a layer to select it, drag to move it, and drag a corner handle or turn the mouse wheel over the selected layer to resize it; arrow keys nudge it. Selection and handles follow the visible part of the layer, so a copied figure's

transparent padding does not get in the way. Press Esc or click the button again to leave the mode; Rasterize also ends it.

Rasterize Image — bakes the placed layers permanently into the image. The layers come off the canvas (they are part of the image now) but stay loaded in their slots, so you can immediately Place the next panel against what you just baked in and rasterize again. This is a single undo step. Markers stay editable. Pasted left/right MW labels that line up with this image's left/right column (for example stacked panels) join its own label sets, so **L/R Values**, **Update Labels** and the font controls update them too; labels that do not line up are added as editable text. Pasted top labels join the top row only if they sit on the same row (otherwise they are skipped and reported). Image settings are reset afterwards.

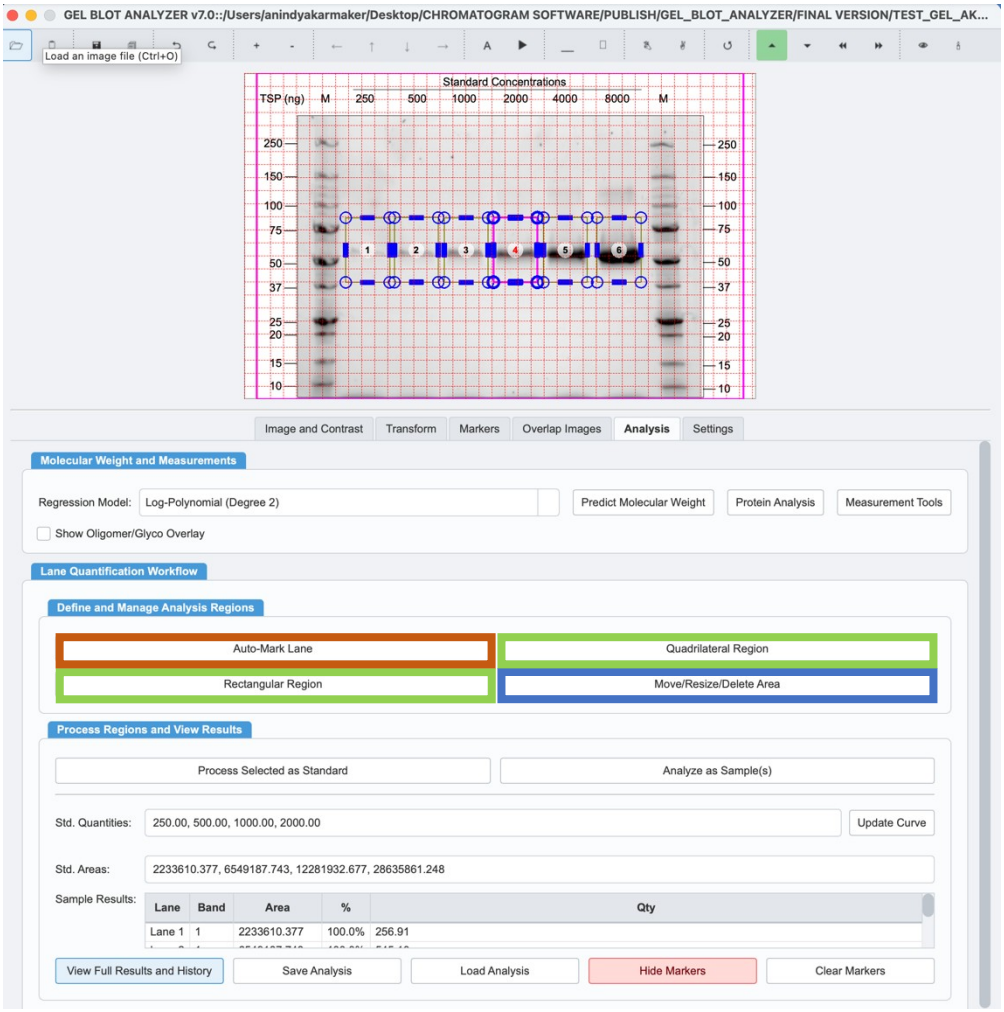
Blend Region / Select Regions to Keep — draw rectangles to restrict blending to chosen areas; Clear All removes them.

Apply Regions To — chooses which layer those regions apply to.

8. Quantitative Analysis Workflow

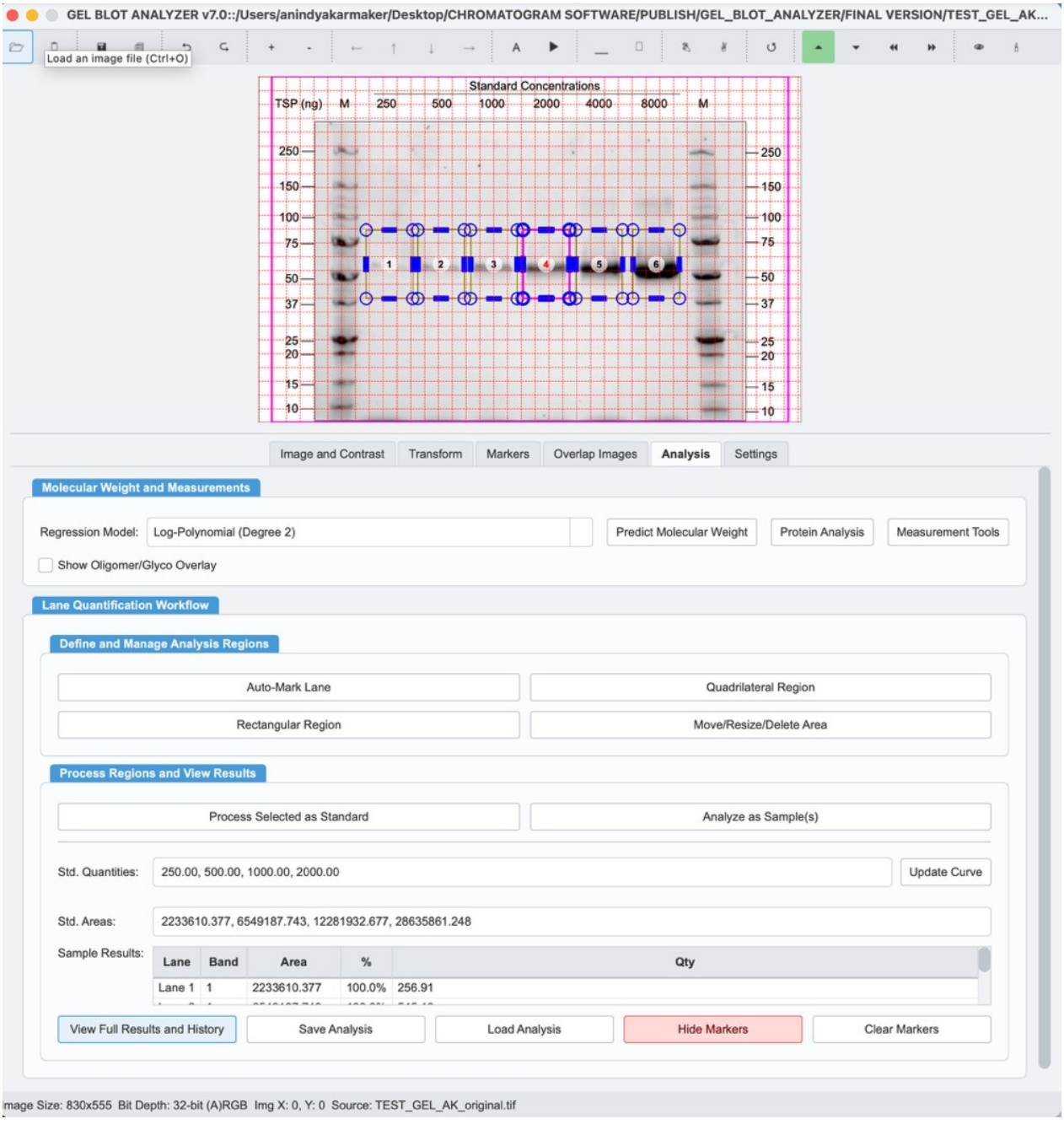
8.1 Defining Analysis Regions

Go to the **Analysis** tab. You must first define the area(s) you want to analyze.



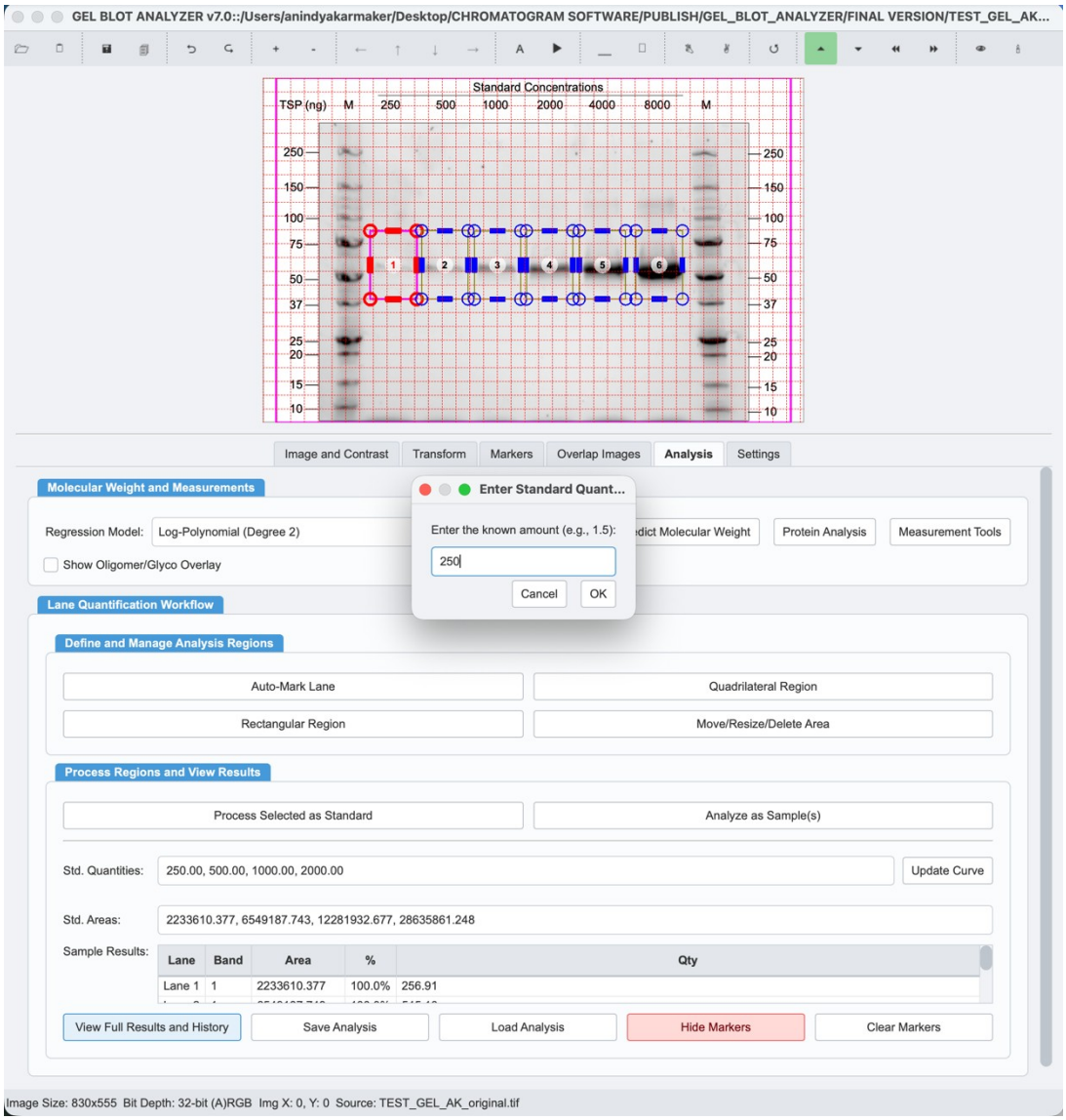
1. For picking region/Lane for analysis:
    - Click **Quadrilateral Region** for skewed lanes. Click four points around your lane starting with top left, top right, bottom right and then final bottom left. **Note:** It is important to follow this direction for proper custom area selection.
    - Click **Rectangular Region** for straight lanes. Click and drag to draw a box around your lane.
    - If you want to use the same size of the bounding box, click on **Auto-Mark Lane** and click on the center of the region around which you want the box. It is a much faster way to create the bounding boxes
2. **Moving/Resizing Regions:** Click **Move/Resize/Delete Area** to enter a mode where you can click a defined lane to select it, then drag it to move or drag its corners to resize.

NOTE: You can use the Snap Grid Functionality to pick uniform areas for analysis since your selection of area impacts the analysis (**CTRL/CMD** + **Shift** + **S**). You can use **Shift key** + **Left Click** Mouse drag to move in straight lines by clicking on the center and you can move edges by clicking on center blue knobs with the same shortcut. You can duplicate the regions by clicking **Ctrl/CMD** + **Shift** + **Left Click** Mouse drag from the center.



8.2 Processing a Standard:

1. Define a region for your band as discussed before.
2. Use the **Move/Resize/Delete Area** button to highlight the region.
3. Click **Process Selected as Standard**.



4. A dialog box opens. Enter the known amount of standard and click **OK**. The Adjust Peak Regions and Calculate Areas dialog box opens. It should detect a single peak. Adjust parameters and click **OK**
5. Repeat for at least two standard lanes to generate a standard curve. The values will appear in the "Std. Quantities" and "Std. Areas" fields. If you want to eliminate custom values, please remove the areas and quantities and click the button **Update Curve**.

Standards can be added before or after the samples are analyzed. Adding or editing a standard recalculates the quantities of samples that have already been analyzed. Their band boundaries and areas are kept, so you don't need to analyze them again. If you

process a lane that has already been analyzed, the standard opens with that lane's bands and settings, and any changes you make are saved back to that lane.

Note: To get the highest precision in quantification, please make sure to use the same parameters for detection particularly the **rolling ball radius**. Also, the quantification is accurately tested up to 4 µg of protein loaded onto the gel.

8.3 Analyzing a Sample:

1. Define the region(s) for your sample lane(s).
2. Click Analyze as Sample(s).
3. The Peak Area Dialog will open for each defined lane. Adjust and click **OK**.
4. The software uses the standard curve to calculate the quantity for each band. The results (areas and calculated quantities) will appear in the **Sample Results** box (resizable; see 4.5).

8.4 Peak Area Dialog

This is where you fine-tune the analysis.



The Profile Plot: Shows the intensity profile along the length of the lane. Detected peaks are marked with 'x', and the calculated area is shaded. The Intensity axis zooms to the profile, so small bands on a high background stay visible. To start the axis at 0 instead, tick **Y-axis from zero** in the bottom row, next to **Persist Settings**. It is off by default and changes only the plot, not the detected bands, areas or quantities.

The Image Strip: A preview of your lane. You can directly click and drag the blue boundary lines to manually adjust the integration area for each peak.

8.5 Global Settings Area Method:

Area Method: Choose the background subtraction method:

Rolling Ball (default, recommended): rolls a ball under the lane profile to remove the gel background, including uneven illumination, gradients and broad smears. Each band is then integrated between its own edges above that background, which gives the most accurate purity (Band %). The **Radius** sets how much background is removed: a smaller radius removes more. The default is about 3% of the lane length; tick **Auto** to size it from the detected band widths. A very large radius left over from an earlier version (larger than 10% of the lane) is replaced by the default automatically, but any radius you set yourself is kept. Rolling Ball is the default for every densitometry and lane/band analysis, including Auto Lane Markers and Auto Analyze Gel. Images saved with earlier versions also open with Rolling Ball for new analyses; their saved results keep the method they were measured with.

Straight Line: one straight baseline is drawn through the valleys (lowest points) between the bands across the whole lane. Fastest; best for clean gels with a flat background.

Rolling-valley: draws a local straight line under each band, between the background values at its two edges. It gives the same results as in earlier versions. On gels with a smeared or humped background it can count background as bands and read purity too low, so prefer Rolling Ball for purity.

Gaussian Deconvolution: Experimental feature. Good for great resolution SDS-PAGE images.

8.6 Manual Peak Adjustment

Use the **Add Peak** and **Delete Peak** buttons to manually correct the detected peaks directly on the plot. **Focus Peak** is used to highlight the peak relevant boundaries for easier manipulation. **Invert Profile Image** is to invert the peaks to troughs and vice-versa for inverted images These buttons sit directly under the graph and stay in place while the settings below them scroll.

8.7 Peak Detection Settings:

Use the sliders (**Min Prominence**, **Min Distance**) to adjust the automatic peak detection algorithm.

Use the **Advanced Detection** feature to tweak the algorithm for peak detection and denoising. By default, the software does not perform denoising and smoothing which may be required for some analysis and that option is there. Good typical values are **0.5** for denoising and **2.0** for smoothing sigma. Use the Min Height and Min Band Width to remove noise or speck from detection. Good starting values are **0 (the default)** and **6 px** respectively.

Detection control reference (tune roughly in this order): the controls below are split between the Basic group and the collapsible Advanced Detection group.

Min Prominence — how far a peak must rise above the dips on either side, as a fraction of the full trace height. This is the single most effective noise control: raise it first if specks are being counted as bands; lower it to recover genuine faint bands.

Min Distance (px) — the minimum spacing allowed between two peaks. Raise it to stop one broad band being split in two; lower it to resolve closely spaced bands.

Band Position — where each band is located along the lane. This position is where molecular-weight marker bands are placed (Auto Lane Markers and Auto Analyze Gel), so it changes the standard curve and predicted molecular weights; band areas are not affected. A magenta tick on the Profile Plot shows the position when it is not the peak maximum. Options:

- **Peak maximum (apex)**: the brightest row of the band (default).
- **Centroid (intensity-weighted mean)**: the band's centre of mass above the background. Sub-pixel; pulled toward a smear.
- **Median (50% of band volume)**: the row that splits the band's volume in half. Less affected by a long tail.
- **FWHM midpoint**: the middle of the band's full width at half maximum; close to where the eye places an asymmetric SDS-PAGE band.
- **Gaussian fit (sub-pixel)**: the centre of a Gaussian fitted to the top half of the band. Best for symmetric bands; not for saturated ones.

Use the same Band Position for the ladder and for any bands you compare against it. The choice is saved with your peak settings and shared with the Auto Gel wizard.

Denoise Sigma — blurs the 2-D image (and widens an impulse-removing median filter) before the trace is built. Raise it for grainy/speckled images; too high merges neighbouring bands. A small median filter always runs, even at 0. Typical: 0.5.

Smoothing Sigma — smooths the 1-D trace only. Helps detection on noisy lanes; too high flattens shoulders and can shift peak centres. Typical: 2.0.

Min Height — the minimum peak height, as a fraction of the trace range. Rejects faint bands outright; Min Prominence is usually the better speck filter. Default: 0 (no height floor, so faint bands are not cut off); raise it only if faint noise is picked up as bands.

Min Band Width (px) — the minimum peak width at half-maximum. Real bands span several pixels whereas single-pixel spikes do not, so raise this to drop sharp noise spikes in the 1-D trace. Typical: 6 px.

Speck Removal (Advanced Detection): this is the most direct control for the common problem of dust, fibres or staining specks being picked up as bands. Unlike **Min Height** and **Min Band Width** (which act on the 1-D trace), **Speck Removal** works on the 2-D lane image before the trace is built: any bright blob smaller than the chosen area (in px²) is erased, so it can never become a peak, while genuine bands — which always span many pixels — are left untouched. Leave it at **0** to disable; start around **10–40 px²** and raise it until the specks disappear.

You can also manually set the number of peaks if the detection algorithm fails by changing the number of detected peaks and click on Set and then narrow the region of peak peak

dragging the blue boundary lines on the **Image Strip**. But the easier option is to just use **Add Peak** to click on the peaks manually. You can also delete the peaks by clicking on it and press delete key or click on the **Delete Peak** button. The **Focus Peak** is useful if you have too many peaks and cannot bind the boundary edges for area estimation. The **Invert Image Profile** is necessary if your bands are white on a dark background (typical for DNA/RNA gels)

Tip — a good starting point: Rolling Ball background with the default radius (or Auto); Min Prominence ≈ 0.05; Min Distance ≈ 10 px; Denoise 0.5; Smoothing 2.0; Min Height 0; Min Band Width 6 px; Speck Removal 0 (raise to 10–40 px² only if specks are still detected). Tune one control at a time and watch the Profile Plot update live; settings you set on one lane are shared with the others, and you can fix individual lanes afterwards.

NOTE: The band detection is tweaked to work only with dark bands on a white background. Please **Invert** the image in the Image and Contrast Setting or use the **Invert Image Profile** in the Peak Detection settings.

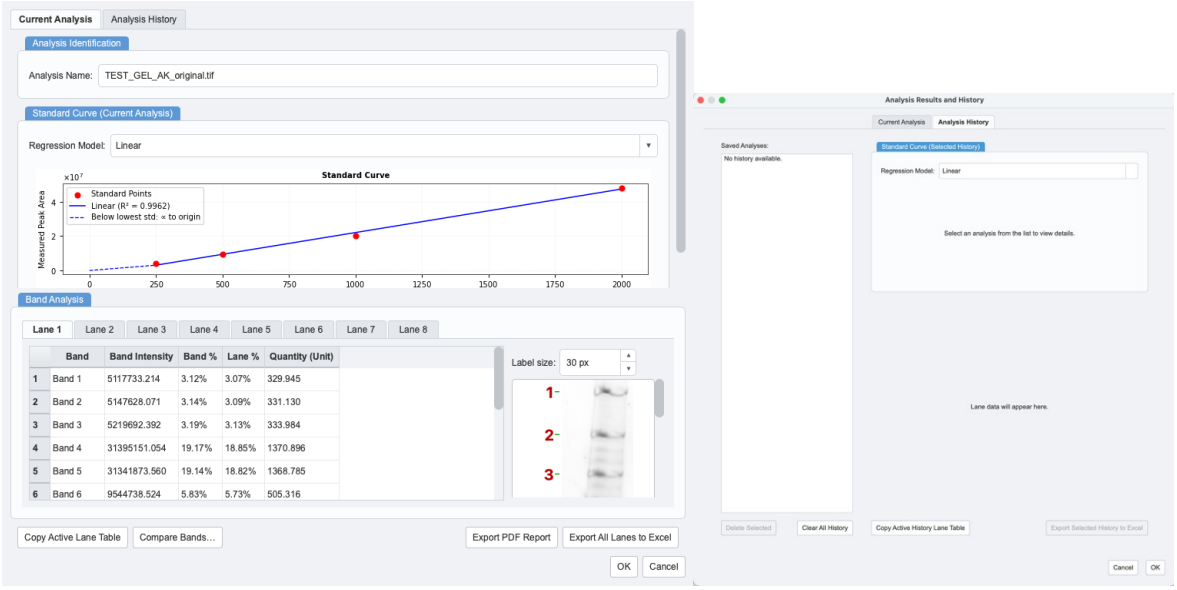
8.8 Copy/Paste Regions:

You can copy the blue boundary lines for analyzing same bands on different regions using **Copy Regions** button and then paste using the **Paste Regions** button. Useful for comparing recovery/quantity. The **Copy Regions** and **Paste Regions** buttons are in the bottom row, next to **OK** and **Cancel**.

Click **OK** to accept the calculated areas and close the dialog.

8.9 Viewing & Exporting Results

Click the **View Full Results and History** button to open the Analysis Results window.



- **Current Analysis** tab: shows a table for each lane and a plot of the standard curve for your most recent analysis. You can copy the table, compare bands, or export all data to an Excel file.
- **Analysis History** tab**:** Every time you click "OK" in this window, the current analysis is saved to a history file. You can browse past results, view their data, and export them to Excel.

The results table lists, for every band:

- **Band Intensity**: the band's background-subtracted volume (the sum of its intensities above the lane background).
- **Band %**: the band's share of all detected bands in the lane. This is the usual measure of purity.
- **Lane %**: the band's share of the whole lane's background-subtracted signal, including signal that is not in a detected band.
- **Quantity**: the amount from the standard curve, when standards are defined. A quantity outside the range of the standards is shown in italics; hover over it for details.

Each lane's table ends with an **All bands** row (the sum of the bands and their share of the lane) and a **Whole lane** row (the lane intensity). The same columns appear in the Excel export and the PDF report. Analyses saved with earlier versions have no lane intensity, so their Lane % shows "—".

Standard curve and Regression Model: when standards are defined, the Current Analysis tab plots them (known amount against measured area) with the fitted curve and its R^2. Choose the curve with the **Regression Model** drop-down: **Linear**, **Power (Log-Log)**, **Polynomial (Deg 2)**, **Polynomial (Deg 3)** or **4-PL Sigmoidal** (needs 4 or more standards). The quantities, the Excel export and the PDF report all use the selected curve, and the choice is saved with the image.

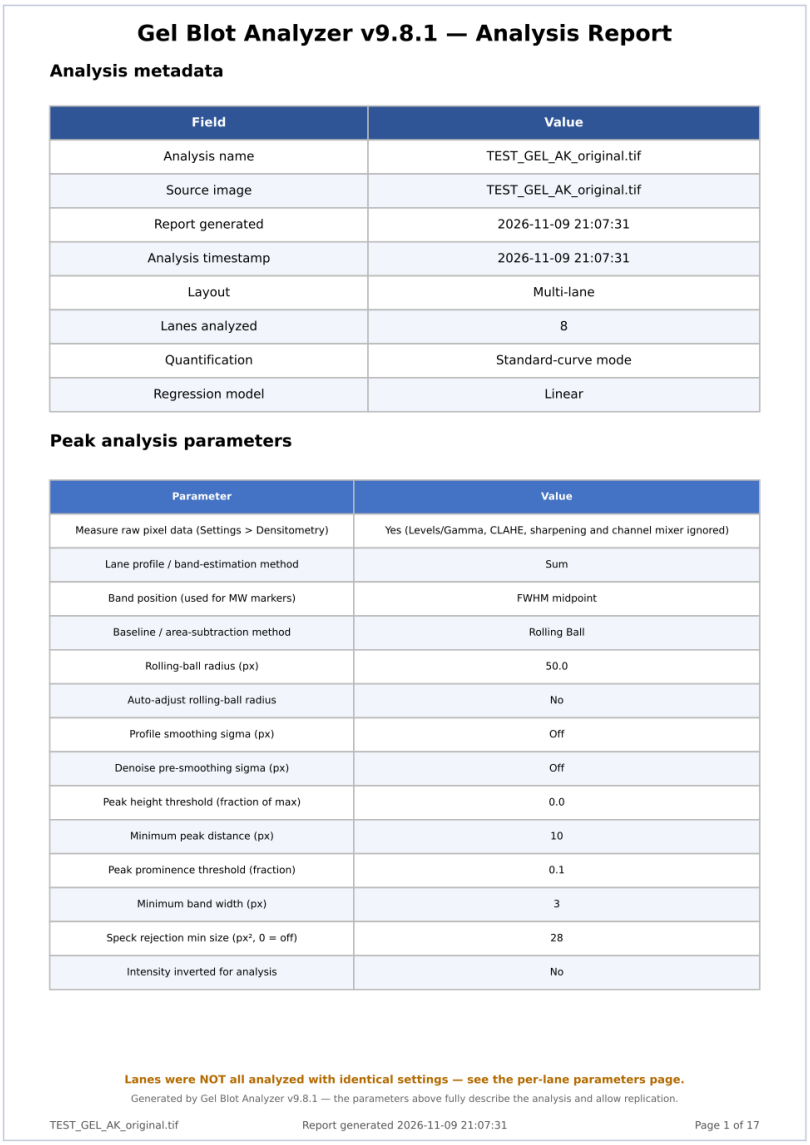
Power (Log-Log) passes through zero and follows standards whose signal levels off at higher loads. It is recommended for faint bands and mass-balance work. With any model, below the lowest standard the amount is taken as proportional to the area (the dashed line from zero on the plot), so quantities are never negative. Above the highest standard, the curve is extended as a straight line. Amounts outside the range of the standards are estimates. For the best accuracy, use standards that bracket your samples.

Compare Bands: click **Compare Bands…** on the Current Analysis or Analysis History tab to compare any band of any lane with any other band, in the same lane or a different one.

- Pick **Lane A / Band A** and **Lane B / Band B**, then click **Add**. Use **Compare All to B** compare every band with one reference band, such as a loading control.
- **Compare by:** Band Intensity, Band %, Lane % or Quantity (when standards are defined).
- Each row shows both values, **A / B**, **A − B**, the **% change** of A relative to B and **log2(A / B)**.
- A pair that is already listed, or a band compared with itself, is not added again. Use **Remove Selected**, **Clear All** and **Copy Table** to manage the list.

- Comparisons are kept while the program is open, saved with the analysis history, and exported to Excel as a **Band Comparisons** sheet with every measure.

Exporting to PDF: The PDF report bundles an annotated image of your gel (with clearly-labelled lane-number badges), the full parameter and method summary (justified for readability), a data-quality / band-saturation check, per-lane band images, and the results tables. The results tables include Band Intensity, Band % and Lane % for every band. The parameter table states whether **Measure raw pixel data** (Settings ▸ Densitometry) was on when the lanes were measured, and the Method summary ends with the purity note: use the Rolling Ball area method and analyse the imager's raw 16-bit export rather than an image exported for publication. Every page has a footer with the image file name, the report date and Page X of N. The report is **protected against editing**: it opens without a password and can be printed and copied from, but PDF readers will not let it be edited, annotated or have its pages rearranged.



Gel Blot Analyzer v9.8.1 — Analysis Report

Analysis metadata

Field	Value
Analysis name	TEST_GEL_AK_original.tif
Source image	TEST_GEL_AK_original.tif
Report generated	2026-11-09 21:07:31
Analysis timestamp	2026-11-09 21:07:31
Layout	Multi-lane
Lanes analyzed	8
Quantification	Standard-curve mode
Regression model	Linear

Peak analysis parameters

Parameter	Value
Measure raw pixel data (Settings > Densitometry)	Yes (Levels/Gamma, CLAHE, sharpening and channel mixer ignored)
Lane profile / band-estimation method	Sum
Band position (used for MW markers)	FWHM midpoint
Baseline / area-subtraction method	Rolling Ball
Rolling-ball radius (px)	50.0
Auto-adjust rolling-ball radius	No
Profile smoothing sigma (px)	Off
Denoise pre-smoothing sigma (px)	Off
Peak height threshold (fraction of max)	0.0
Minimum peak distance (px)	10
Peak prominence threshold (fraction)	0.1
Minimum band width (px)	3
Speck rejection min size (px², 0 = off)	28
Intensity inverted for analysis	No

Lanes were NOT all analyzed with identical settings — see the per-lane parameters page.

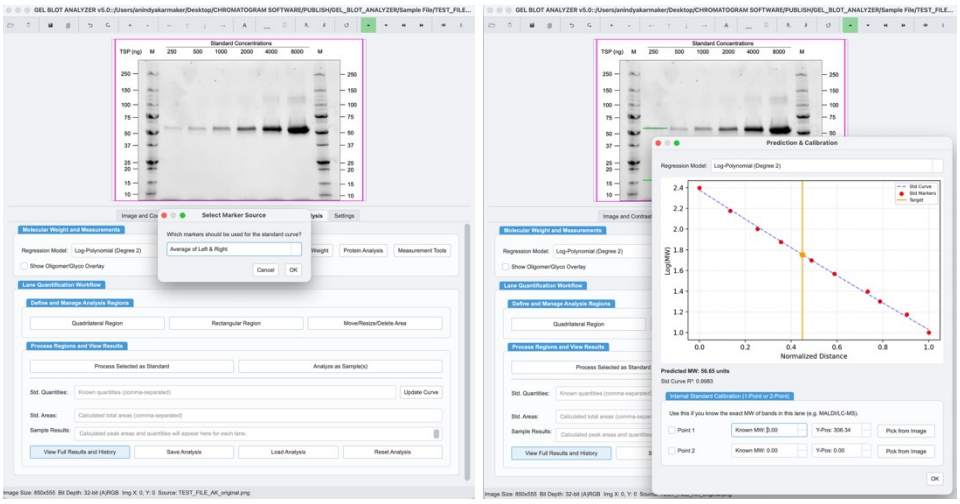
Generated by Gel Blot Analyzer v9.8.1 — the parameters above fully describe the analysis and allow replication.

TEST_GEL_AK_original.tif Report generated 2026-11-09 21:07:31 Page 1 of 17

First page of the PDF report: the analysis metadata and every peak-detection parameter, so the result can be reproduced.

9. Molecular Weight Prediction

This feature allows you to estimate the size of a protein band based on your standard markers.



9.1 Setting up the Standard Curve

1. Place your standard markers using the tools in the **Markers** tab. You need at least two numeric markers on either the Left or Right side.
2. Go to the **Analysis** tab.
3. Click the **Predict Molecular Weight** button (**Ctrl**/**CMD** + P).
4. If you have markers on both sides, you will be asked which set to use (Left, Right, or an Average). With **Average**, each marker value found on both sides is placed halfway between its left and right positions, which evens out a gel that ran slightly faster on one side. Values found on only one side are left out.
5. The cursor will change. Click on the center of the band you want to analyze.

Tip: marker bands placed automatically (Auto Lane Markers, Auto Analyze Gel) sit at the **Band Position** chosen in Peak Detection (see 8.7). Click your sample band at the matching point (for example its centre for FWHM midpoint or Centroid, its darkest row for Peak maximum) so both are measured the same way.

6. The **Prediction and Calibration** dialog appears.

This window shows a plot of your standard markers (log scale) and the regression line used for the calculation. The predicted molecular weight and the R^2 value (goodness of fit) are displayed. You can change the Regression Model to see which one best fits your standards. The available models are Log-Linear (Degree 1), Log-Polynomial (Degree 2), Log-Polynomial (Degree 3), and Log 4-PL.

NOTE: Usually **Log-Polynomial (Degree 2)** fits best for a gradient gel for accurate estimation without overfitting.

Internal Standard Calibration: if you know the true mass of one or two bands (for example from MALDI or LC-MS), tick **Point 1** (and **Point 2**), enter the **Known MW** and use **Pick from Image** to click that band. One point shifts the whole curve so it passes through the known mass; two points also correct its slope. The calibrated curve is drawn in green and the result is marked [Calibrated: 1-Pt] or [Calibrated: 2-Pt].

Changing the model later: the **Regression Model** drop-down on the Analysis tab refits the same ladder with the new model and keeps your calibration points. The predicted MW and any oligomer/glyco callouts update straight away, without picking the band again.

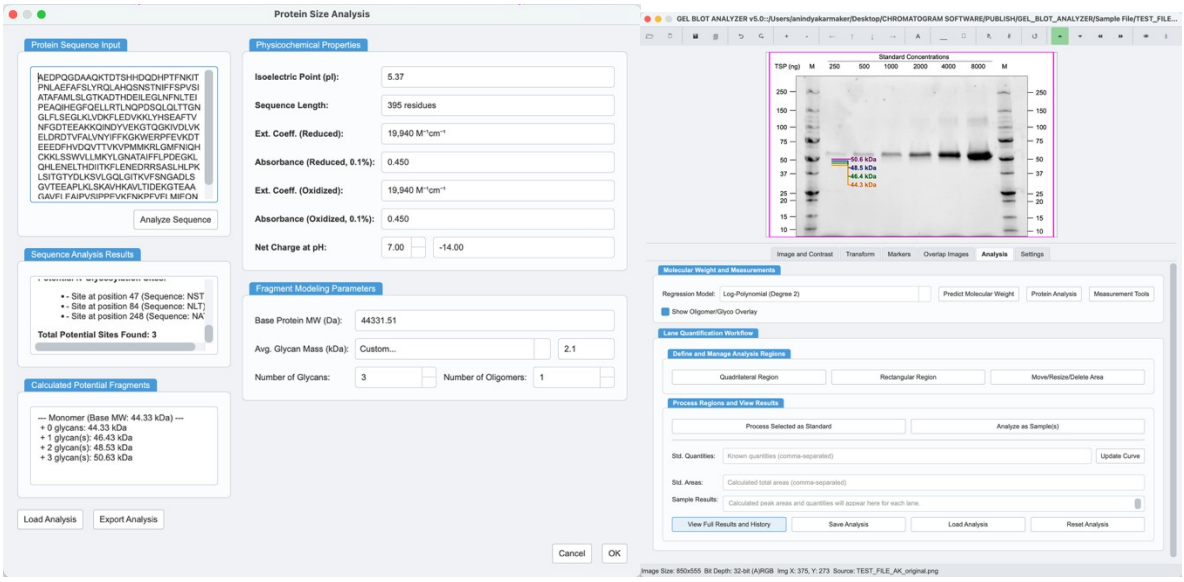
Click **OK** to accept the prediction.

If **Show Oligomer/Glyco Overlay** (Analysis tab) is ticked, the oligomer callouts appear as soon as you accept the prediction — no Protein Analysis is needed (see 10.1).

10. Advanced Features

10.1 Protein Size Analysis (Oligomers & Glycosylation)

This tool helps you hypothesize the identity of bands based on a protein's primary sequence and potential modifications.



1. Click the **Protein Analysis** button on the **Analysis** tab.
2. Paste your protein's amino acid sequence into the top text box.
3. Click the **Analyze Sequence** button. This will automatically calculate the theoretical **Molecular Weight (MW)** and populate the "Base Protein MW (kDa)" field. Moreover, it will calculate the **Molar Extinction Coefficient** (for both reduced and oxidized forms) based on Tryptophan, Tyrosine, and Cysteine content, identify potential **N-Glycosylation sites** (N-X-S/T sequons) and display a formatted sequence view with residue numbering and a summary of the analysis.
4. In the dialog, you can now fine-tune the model: **Base MW:** The calculated value can be manually overridden if you have a more accurate experimental value. **Avg. Glycan Mass:** Defaults to **0** (no glycosylation); a blank field also counts as 0, so the oligomer masses are always calculated. Select a common glycan type from the dropdown menu to auto-fill the mass, or select "Custom..." to enter your own value (e.g., 2.5 kDa). **Number of Glycans:** This defaults to the number of potential sites found but can be changed to model different glycosylation states. If you set it above 0 while no glycan type is chosen, the first glycan type (N-linked High-Mannose, Man5, 1.2 kDa) is selected and its mass filled in automatically. **Number of Oligomers:** Set the number of oligomeric states to model (e.g., 2 for dimer). The **Calculated Potential Fragments** box will automatically update to list the theoretical molecular weights for all combinations of the parameters you have set.
5. You can save or load your entire analysis using the buttons at the bottom:

- **Export Analysis:** Saves all the information (sequence, parameters, results) to a readable text file.
- **Load Analysis:** Loads a previously saved analysis file, repopulating all fields in the dialog.

6. After clicking **OK**, tick **Show Oligomer/Glyco Overlay**. If a molecular weight prediction already exists the callouts appear immediately; otherwise they appear as soon as you accept a **Predict Molecular Weight** result.

Show Oligomer/Glyco Overlay: marks the predicted masses of every modeled oligomer and glycosylation state on your gel. Each band gets a callout: a colored arrow on the band and a matching label showing its mass in kDa. If no protein size or glycan data has been entered in Protein Analysis, the overlay uses the current **Predict Molecular Weight** result as the monomer (no glycans). Clearing the Protein Analysis data switches back to that, and every new prediction updates the callouts immediately.

Masses beyond the ladder: a callout is placed even when its mass is larger or smaller than every marker. Beyond the part of the curve the ladder covers well, the curve is continued as a straight line, so this works with every regression model, including Log 4-PL, which flattens out at the ends. The status bar lists these masses as extrapolated; treat their positions as approximate.

Editing callouts: each label is an ordinary custom marker and each arrow a linked line, so callouts can be edited in **Move/Resize**, are included in Undo, and are saved, copied and exported (including the editable PowerPoint export). The arrow tip stays on its predicted band; drag the label and the line follows it. Put the label on the other side of the band and the arrow turns to point from that side. Double-click a label to change its text, font or color; deleting a label also deletes its line. Labels you have moved keep their position when the prediction or the partial-glycosylation option changes. Un-tick the box to remove all callouts.

Staying in view: callouts are laid out inside the preview area (at most 15% beyond it), and the preview zooms out automatically so every label and arrow is visible. If labels sit outside the image, use **Set Recommended Values** and **Apply Padding** in the Transform tab before saving.

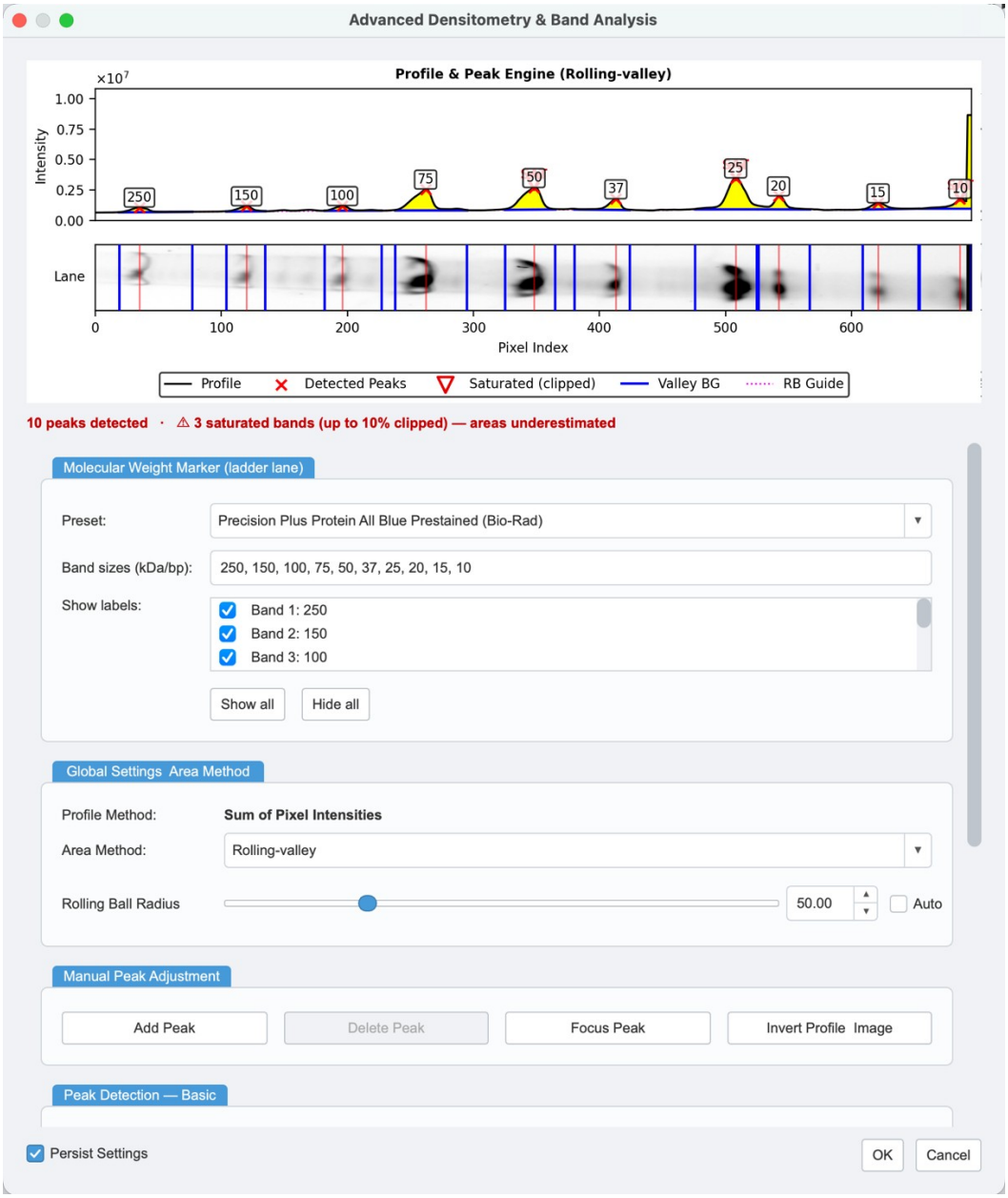
The labels are grouped by oligomer: every form of one oligomer (its base mass plus all of its glycosylated variants) shares one aligned column, and each successive oligomer group steps to the right in a clean cascade that begins at the end of the previous group, so labels stay readable even when many bands are close together.

Show Partial Glycosylation: This checkbox controls how glycosylation states are listed. When it is OFF, only the fully-glycosylated form of each oligomer is shown. When it is ON, every partial-occupancy band is shown, including the unglycosylated backbone. For example, a 50 kDa protein with three 2 kDa glycan sites shows 50, 52, 54 and 56 kDa with the option ON, and only 56 kDa with it OFF.

For oligomers, the number of glycosylation sites scales with the number of subunits — a dimer of a protein with 3 sites has 6 sites in total — so both its fully-glycosylated mass and its partial ladder are calculated accordingly.

10.2 Automatic Lane Detection

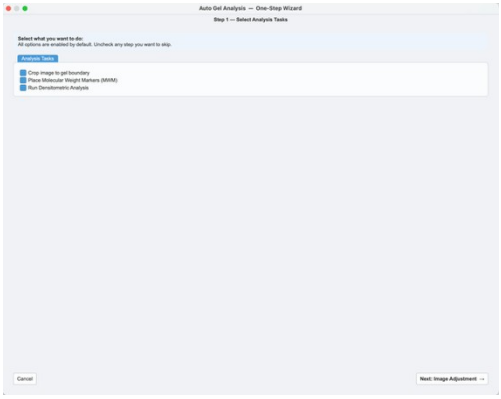
This feature helps you quickly place markers for a set of straight or skewed lanes.

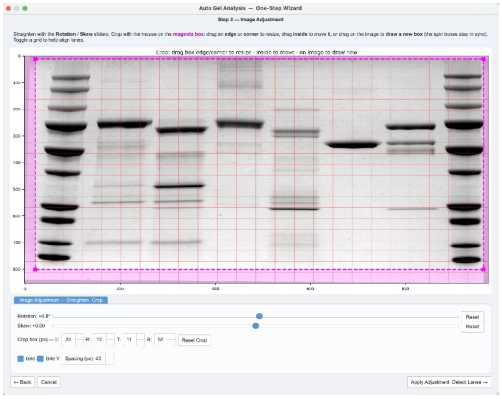


1. Click the **Automatic Lane Markers** button on the toolbar.
2. **Select Marker Side** (Left or Right) to place the markers.
3. **Select Lane Region Type** ("Rectangle" for straight lanes, "Quadrilateral" for skewed).
4. Draw the region covering all the bands in your lane.
5. The **Advanced Densitometry & Band Analysis** dialog will open, showing the intensity profile. Adjust the sliders to ensure all your bands are detected as peaks ('x' markers). Use the delete key to delete unintended peaks

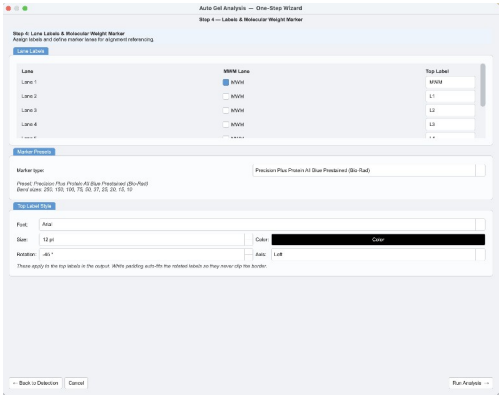
6. You can select the default Preset options or use the **Custom** preset to put the band sizes. You can also hide band sizes by unchecking from **Show labels**.
7. Click **OK**. The software will place the standard markers at each detected band, using the **Band Position** chosen in Peak Detection (see 8.7).

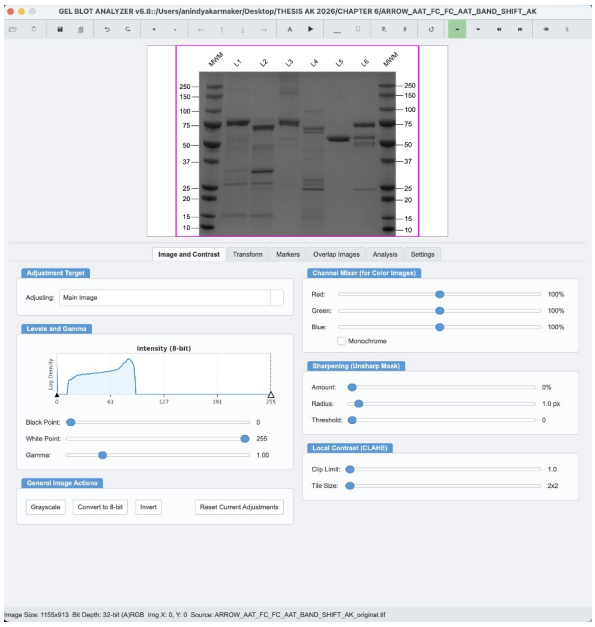
10.3 Automated One-Step Gel Analysis (Auto Analyze Gel)









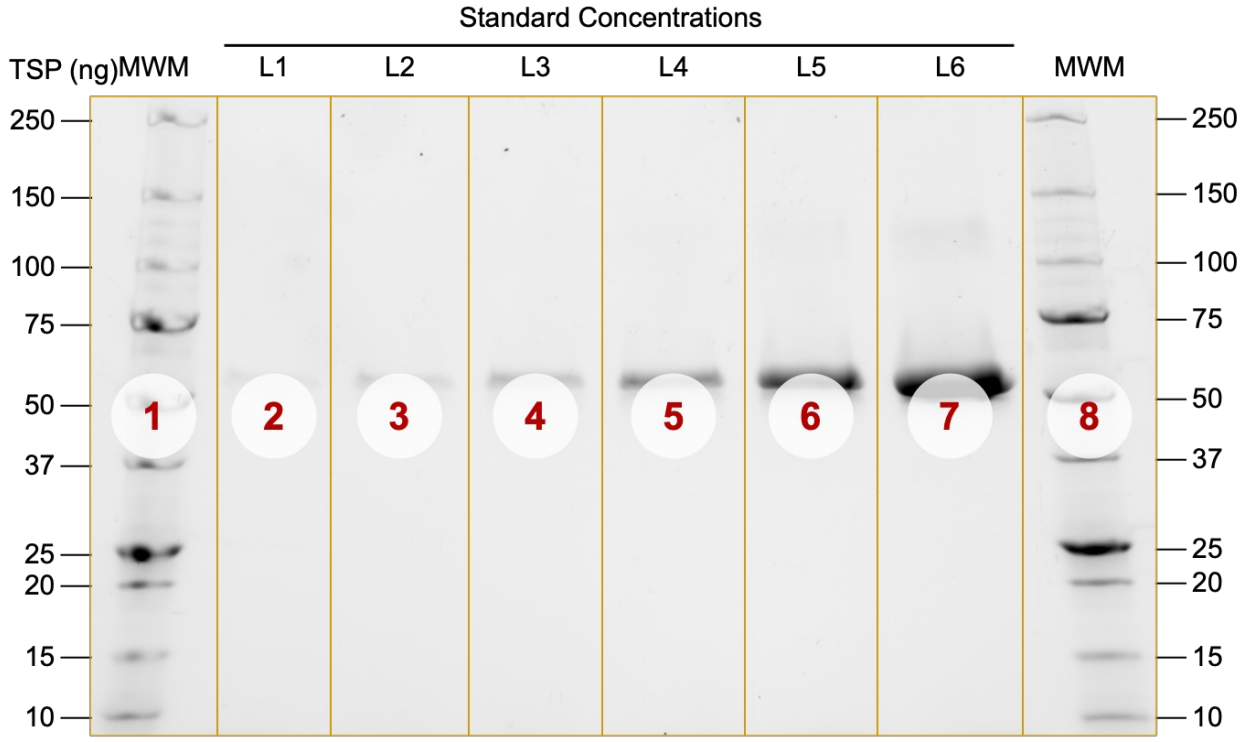


The Auto Analyze Gel tool runs an entire workflow—from a raw image to densitometry results—through a single guided wizard. Launch it from the “Auto Analyze Gel” button on the toolbar or from the Tools menu. It is ideal for quickly quantifying an entire gel without manually defining each lane.

When you start the tool, you are first asked how to draw the gel boundary: choose Rectangle for a straight gel or Quadrilateral for a skewed/tilted gel, then draw the region around all of your lanes. The One-Step Wizard then walks you through five phases:

1. **Task Selection:** Choose whether to run full densitometry with quantification or a marker/labeling-only pass.
2. **Image Adjustment:** Straighten the gel with the Rotation and Skew sliders, fine-tune the crop with a draggable crop box, and toggle a grid for alignment. Keyboard: ← / → change the rotation by 0.1° (hold Shift for 1°) and ↑ / ↓ change the skew by 0.01 (Shift: 0.10), even while the preview has focus; the Rotation and Skew readouts have a fixed width so the sliders do not jump.
3. **Lane Detection:** The software builds a column-intensity profile and automatically places lane boundaries on the prepared canvas. Drag the boundary lines to add, remove, or reposition lanes. **Resize the lanes first** and then add missing lanes. It will number then accurately. To trace lanes that run differently (smiling, bending or shorter lanes), click a lane to select it: drag its **corner handles** to reshape it, its **top or bottom edge** to make it shorter or taller, or **inside it** to move it. **Reset Lane Shape** makes the selected lane a full-height upright lane again. A reshaped lane is straightened before analysis, so densitometry and marker positions follow its outline, and it is added to the main window as a quadrilateral lane. Moving a lane-detection slider re-detects the lanes and clears any reshaping.
4. **Labels & Molecular Weight lane:** Pick the molecular weight marker lane and rename the lane top-labels
5. Band Detection & Analysis Type: Adjust the molecular weight marker options and adjust the band/peak detection settings. For peaks not detected, can manually add them or adjust the height/prominence settings **Min Height** starts at **0** so faint bands are not cut off; raise **Min Prominence** if specks are detected. Choose the **Band Position** method (see 8.7) here too; the wizard uses FWHM midpoint until you choose another.

When you finish, the wizard applies all geometry (rotation, skew, crop, padding) to a full-color, full-resolution copy of your image, places the markers, runs densitometry on every detected lane, and opens the Analysis Results window with the combined data.



A gel after Auto Analyze Gel: padding and lane labels added, the ladder lanes labeled on both sides, and the numbered densitometry regions ready for the results window.

The molecular-weight ladder starts from your current marker values — including values you edited or deleted in the main window — and any changes you make to the ladder or its bands in the band dialog are applied to the image. Existing left/right MW labels and custom markers are kept and moved onto the cropped gel, and the final padding is measured with all of them so nothing is cut off. The wizard's plots re-fit their labels whenever you resize the window.

10.4 Measurement Tools

The Measurement Tools window—opened with the Measurement Tools button on the Analysis tab—provides image calibration and on-image measurements in a separate, always-on-top window, so you can keep working in the main viewer. It offers three tools:



1. Set Scale: Calibrate the image by drawing a line across an object of known length (e.g., a ruler or printed scale bar) and entering its real-world length. The active scale is shown in the window.

2. Measure Distance: Click two points to measure the straight-line distance between them. The result is reported in calibrated units once a scale is set, otherwise in pixels.
3. Measure Area: Click and drag to draw a freehand shape and calculate its enclosed area.

   Use Clear All to remove the current scale and any measurement overlays. Closing the window deactivates the active measurement tool.

10.5 Application Settings (Settings Tab)

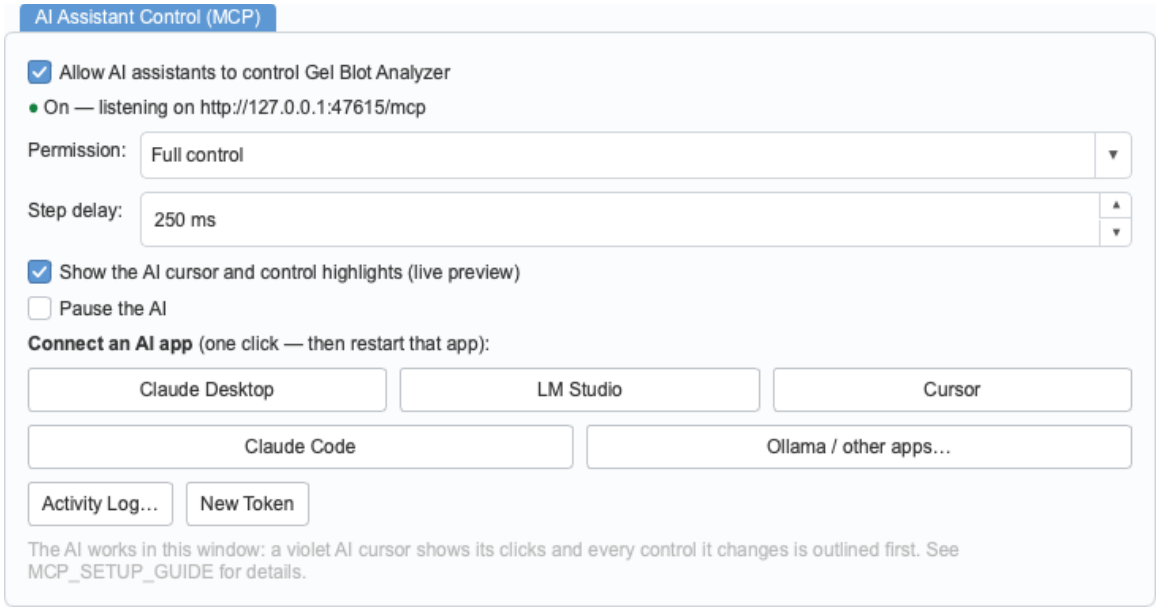
The Settings tab (the sixth tab) holds application-wide preferences. Click Apply and Save Settings to store them. Some options take effect immediately, while others require restarting the program.

1. **Processing Unit Priority:** Choose whether image processing runs on the GPU or the CPU. When GPU is selected you can pick the target OpenCL device (GPU:0 is typically the integrated GPU, GPU:1 the discrete GPU). A live label shows the currently active device. Switching CPU/GPU applies immediately; changing the specific GPU ID requires a restart.
2. **User Interface Scaling:** Adjust the overall UI scale (50%–250%) to suit high-DPI or small screens. Requires a restart.
3. **Application Dimensions:** Set the image-viewer width and height and the program UI width (in pixels); changes take effect after a restart. The viewer now scales its content to these dimensions — the image, markers and labels all zoom together to fill the viewer. Larger values simply give a bigger, proportionally-scaled view.
4. **Autosave & Crash Recovery: Autosave** sets how often a recovery copy is saved: Off, or every 30 seconds up to every 30 minutes (default: every 1 minute). **Autosave on crash** (on by default) also saves a copy the moment an unexpected error occurs, so changes made since the last timed autosave are kept. **Open Recovery Folder** shows where the copies are kept. See 3.3.
5. **Densitometry ▸ Measure raw pixel data** (on by default, recommended): band volumes are measured on the recorded pixel intensities, so Levels/Gamma, CLAHE, sharpening and the channel mixer change only how the image looks, not your results. Inversion still applies. Untick it only to reproduce results from versions before 9.6, which measured the display-adjusted image. The PDF report records this setting for every analysis and includes the note shown under it in its Method summary.
6. **AI Assistant Control (MCP):** lets an AI assistant operate GBA. Off by default; see 10.6.
7. **Reset All Settings:** Restores all application preferences (viewer size, UI scale, CPU/GPU choice and saved marker presets) to their defaults by deleting the saved configuration file. You are asked to confirm first. This does NOT affect your per-image saved configuration files. A restart is recommended afterwards.

10.6 AI Assistant Control (MCP)

GBA can be operated by an AI assistant, such as Claude, LM Studio, Cursor, VS Code or a local Ollama model. It uses the **Model Context Protocol (MCP)**, the open standard these apps use to call tools. The AI works live in your window: a violet **AI** cursor shows each click and drag, and every control is outlined just before the AI changes it, so you can watch, pause or stop it at any time.

AI control is OFF by default. Nothing can connect until you turn it on yourself, and turning it off disconnects every AI immediately. Full connection details for every app are in **MCP_SETUP_GUIDE.md**, which ships with the program.



The AI Assistant Control group on the Settings tab. The status line shows the address an AI app connects to.

Turning it on

1. Open **Settings ▸ AI Assistant Control (MCP)** (or **Tools ▸ AI Assistant Control (MCP)…**, which jumps to the same group).
2. Tick **Allow AI assistants to control Gel Blot Analyzer**. The status line changes to "● On — listening on http://127.0.0.1:47615/mcp".
3. Click the button for your AI app, then restart that app. Ask it, for example, "Use Gel Blot Analyzer to show me the open image."

Button	What it does
Claude Desktop	Adds GBA to Claude Desktop's settings automatically.
LM Studio	Adds GBA to LM Studio's mcp.json automatically.
Cursor	Adds GBA to Cursor's mcp.json automatically.
Claude Code	Copies a ready-made "claude mcp add …" command to paste into a terminal.

Button	What it does
Ollama / other apps…	Shows the connection details (command, URL and token) for any other app, including Open WebUI and AnythingLLM.

Your existing entries in those files are kept, a one-time backup is saved next to each file, and a file GBA can't read is never overwritten. If an AI app needs GBA while it is closed, GBA starts automatically, but only when AI control is turned on.

AI settings

Setting	What it does
Allow AI assistants…	Master switch. Off by default.
Permission	**Full control**, or **View only** (the AI can look and read results but can't change anything).
Step delay	Pause before each click or change (default 250 ms) so you can follow along. 0 = full speed.
Show the AI cursor and control highlights	Shows the violet AI pointer and the control outlines.
Pause the AI	Refuses all AI requests until you untick it.
Activity Log…	A live, timestamped list of every AI action and its result (also under **Tools ▸ AI Activity Log…**).
New Token	Revokes access for apps connected by URL. One-click connections keep working.

The AI can never see or change these settings, and some confirmations, such as **Reset Image**, can only be answered by you.



The Activity Log lists every tool the AI calls, how long it took and its result. Use **Pause the AI** here to stop it at any moment.

Smart gel tools (new in 9.8)

These tools do a whole job the way an expert would, using GBA's own analysis code. They are designed so that even small local models analyze gels and blots correctly:

Tool	What it does
`inspect_gel`	Reads the gel first: whether it must be inverted, the lanes, the real bands in each lane, the ladder lane, empty lanes and possible standards. It returns a short summary and the questions the AI should ask you.
`fix_band_polarity`	Inverts the image only when bands are light on a dark background (DNA/RNA gels, stain-free, fluorescence, chemiluminescence) so lanes are white and bands black. It never inverts twice.
`label_mw_ladder`	Labels the molecular-weight ladder with **Auto Lane** analysis (10.2), then adds padding for the labels.
`auto_analyze_gel`	Runs **Auto Analyze Gel** (10.3) from start to finish: lanes, lane labels, ladder MW labels and densitometry.
`quantify_with_standards`	Quantifies every band against standard lanes whose amounts you give, and reports the curve fit (R^2).

Tool	What it does
`combine_images`	Places a western blot below, above, left or right of the gel, adds the padding and matches the width or height (see Chapter 7).
`get_workflow_guide`	Step-by-step recipes for protein gels, DNA gels, western blots, densitometry, standards and combining images.

Every other button, slider, checkbox, list and text field in every window is also reachable, along with menu commands, dialogs, saving and PowerPoint export.

How the AI analyzes a gel

1. **Inspect first.** The AI calls inspect_gel and asks you anything it can't be sure of, for example the number of lanes, which lane is the ladder, and which ladder (preset) you used. Lanes that contain only one faint band each can be hard to separate, so confirm the lane count when asked.
2. **Fix the polarity.** DNA/RNA gels and fluorescent or chemiluminescent images are inverted so bands are black on white, which densitometry needs.
3. **Label the ladder.** For figure labels, the ladder is labeled with Auto Lane analysis. If the ladder shows more or fewer bands than the preset, the AI picks the bands that fit a smooth MW curve and asks you to confirm.
4. **Densitometry.** When you ask for quantification, Auto Analyze Gel is used. Only true bands are kept. A band must be at least 5× the lane's background noise, at least 2 px wide, and at least 1 % of the strongest band in its lane and on the gel. Background bumps and noise are discarded.
5. **Standards.** When neighboring lanes show a band of the same size with rising or falling intensity, the AI asks whether they are standards and how much each contains. It never guesses amounts. With your amounts it fits a standard curve (Linear, Power, Polynomial or 4-PL; see 8.2 and 8.9) and calculates the samples. If the fit is poor (R^2 below 0.95), for example because the top standard is saturated, it warns you and suggests another model.
6. **Combine and save.** Optionally the blot is placed with the gel, then the image and its configuration are saved.

Example requests

- "Analyze the open gel. It's an SDS-PAGE gel with the Precision Plus ladder. Label the ladder and run densitometry."
- "This is an agarose DNA gel with a 1 kb Plus ladder in lane 1. Make the bands black on white and label the ladder."
- "Lanes 2–6 are BSA standards: 125, 250, 500, 1000 and 2000 ng. Quantify the other lanes."
- "Open ~/Blots/actin_wb.tif and put it under the gel with a 30 px gap." or "…put it to the right of the gel."

Using a local model (Ollama)

Ollama only runs models; its own chat window can't use tools. Connect it through a front-end that supports tools: **AnythingLLM** (choose Ollama as the provider, add GBA's MCP entry, and start messages with @agent), **Open WebUI** (add GBA as a tool server at http://127.0.0.1:47615 with the token), or another MCP client such as mcphost or Continue. Use a general model with reliable tool calling, for example qwen3:8b, qwen2.5:7b (instruct), llama3.1:8b or mistral-small. Coder models (such as qwen2.5-coder) write the tool call as text instead of calling it, and reasoning models (such as deepseek-r1) are very slow. If nothing seems to happen, open **Tools ▸ AI Activity Log…**: if no action appears, the model or front-end is not calling the tools.

Security

- GBA listens on this computer only (127.0.0.1), and every request needs a secret token.
- Web pages from other sites can't connect. Quitting GBA or turning the switch off ends all AI access.
- Use **View only** when you only want advice.

11. Keyboard Shortcuts

Action	Windows/Linux	macOS
File		
Load Image	Ctrl+O	Cmd+O
Save Image & Config	Ctrl+S	Cmd+S
Edit		
Undo	Ctrl+Z	Cmd+Z
Redo	Ctrl+Y	Cmd+Shift+Z
Copy Rendered Image	Ctrl+C	Cmd+C
Paste Image	Ctrl+V	Cmd+V
View		
Zoom In	Ctrl+=	Cmd+=
Zoom Out	Ctrl+-	Cmd+-
Pan View	Arrow Keys	Arrow Keys
Tools & Modes		
Reset Image & Annotations	Ctrl+R	Cmd+R
Predict Molecular Weight	Ctrl+P	Cmd+P
Clear Analysis Data	Ctrl+Shift+P	Cmd+Shift+P
Place Left Marker	Ctrl+Shift+L	Cmd+Shift+L
Place Right Marker	Ctrl+Shift+R	Cmd+Shift+R
Place Top Marker	Ctrl+Shift+T	Cmd+Shift+T
Place Arrow Marker	Ctrl + Arrow	Cmd + Arrow
Move/Resize Items (toggle)	Ctrl+Shift+M	Cmd+Shift+M
Toggle Grid and Snap (X & Y)	Ctrl+Shift+N	Cmd+Shift+N

Toggle Line X	Ctrl+Shift+X	Cmd+Shift+X
Toggle Line Y	Ctrl+Shift+Y	Cmd+Shift+Y
Increase Grid Size	Ctrl+Shift+ ↑	Cmd+Shift+ ↑
Decrease Grid Size	Ctrl+Shift+ ↓	Cmd+Shift+ ↓
Toggle Guide Lines	Ctrl+G	Cmd+G
Convert to Grayscale	Ctrl+B	Cmd+B
Invert Image	Ctrl+I	Cmd+I
Toggle Snap to Grid	Ctrl+Shift+N	Cmd+Shift+N
Toggle Dark/Light Mode	Ctrl+Shift+D	Cmd+Shift+D
Auto Gel: Rotation (←/→), Skew (↑/↓); Shift = larger step	Arrow keys	Arrow keys
Move/Resize: nudge selected item (Shift = 10 px)	Arrow keys	Arrow keys
Move/Resize: add/remove item from selection	Shift+Click	Shift+Click
Move/Resize: delete selected item(s)	Delete	Delete
Quickly Switch Tab	Ctrl+1/2/3/4/5/6,	Cmd+1/2/3/4/5/6,
Exit Any Active Mode	Esc	Esc

12. Troubleshooting

- **"A required library is missing" error on startup:** Make sure you have installed all the dependencies listed in the Installation section.
- **Molecular Weight Prediction is grayed out:** You need to place at least two standard markers on the Left or Right side before this option becomes available.
- **Image looks distorted after rotation or skew:** These operations are previews. Click the "Apply" button in the **Transform** tab to make the change permanent and see a high-quality result.
- **Cannot move or resize a lane area:** Make sure you have clicked the **Move/Resize Selected Area** button on the **Analysis** tab to enter the correct mode.
- **The application crashes unexpectedly:** An error_log.txt file is created in the same folder as the application (next to the executable). It records the full error details for every kind of failure — including errors raised inside the interface, in background tasks, and low-level (native) crashes — so it is the first place to look when reporting a problem. Each unexpected-error dialog also has an expandable Details section containing the same traceback. With autosave on (see 3.3), the next launch offers to restore your unsaved work.
- **Purity (Band %) seems too low:** use the **Rolling Ball** area method and keep **Settings ▸ Densitometry ▸ Measure raw pixel data** on. If faint background bumps are detected as bands, raise **Min Prominence** or **Min Height**, or delete those peaks. For absolute intensities, analyse the imager's raw 16-bit export rather than an image exported for publication.
- **A saved JPG has a white background instead of transparency:** JPEG cannot store transparency. Save as PNG or TIFF instead (GBA offers this when you choose JPG).
- **“Could not save the configuration file … Object of type ndarray is not JSON serializable”:** fixed in 9.8. In earlier versions, reopening an image that had a molecular-weight prediction, or changing the Regression Model, made every later config save of that image fail. The image files themselves were saved. Open the image in 9.8 and save it again to write its configuration. If a save ever fails because of the data rather than the folder, the message now says so.
- **“Could not save the configuration file” or “…cannot be saved” in Box, Google Drive, OneDrive, Dropbox or Downloads:** these folders can lock a file while they upload or scan it. Wait a moment and save again, pause syncing while you save, or save to a local folder (e.g. Documents) and copy the files across afterwards.
- **The AI assistant can’t connect or does nothing:** check that **Settings ▸ AI Assistant Control (MCP)** is on and not paused, restart the AI app after clicking its connect button, and open **Tools ▸ AI Activity Log…** to see whether any request arrives. Ollama needs a tool-capable front-end and model (see 10.6). An AI app that launches an older installed copy of GBA won’t have the 9.8 smart tools.

- **The Analysis Results window shows an older analysis:** fixed in 9.8.1. Previously, if the results window was still open from an earlier analysis (or an earlier image), a new analysis or **View Full Results and History** kept showing the old table. The window now updates to the current results; close it or press OK if you prefer to start from a fresh one.
- **The AI reports “Gel Blot Analyzer did not become ready within 90s” although GBA is open:** fixed in 9.8.1. This happened when one copy of GBA quit while another started (for example when the AI app launched a second copy) and the quitting copy erased the new copy’s connection details. GBA no longer overwrites another running copy’s details, and the connector now finds the running copy by itself even if the recorded port is stale. If it still happens, quit every copy of GBA, open it once, and retry.
- **“cannot import name 'cwt' from 'scipy.signal'” when running from source:** SciPy 1.15 and later removed this function. Fixed in 9.8.1 (it was never used); earlier versions need SciPy 1.13 or 1.14.